

9. fejezet

A Cisco Adaptive Security Appliance

CCNA Security v2.0



Tartalom

9.1 Bevezetés az ASA-ba

9.2 ASA tűzfal beállítása

9.1

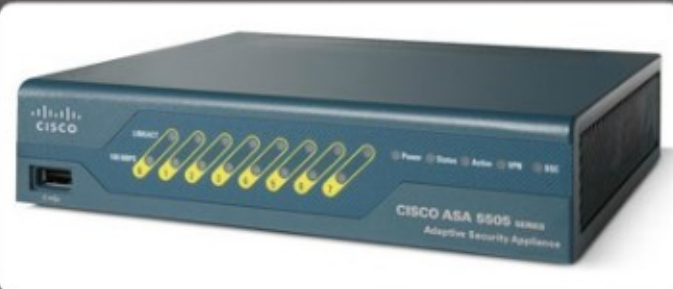
Bevezetés az ASA-ba

9.1.1 ASA megoldások



ASA tűzfal modellek

Small Office and Branch Office ASA modellek

	ASA 5505 / Security Plus	Up to 150 Mbps
	ASA 5506-X/Security Plus	750 Mbps
	ASA 5512-X/Security Plus ASA 5515-X	1 Gbps 1.2 Gbps

ASA tűzfal modellek

Internet Edge modell



ASA tűzfal modellek

Enterprise Data Center modellek



ASA 5585-X SSP10

4 Gbps

ASA 5585-X SSP20

10 Gbps

ASA 5585-X SSP40

20 Gbps

ASA 5585-X SSP60

40 Gbps

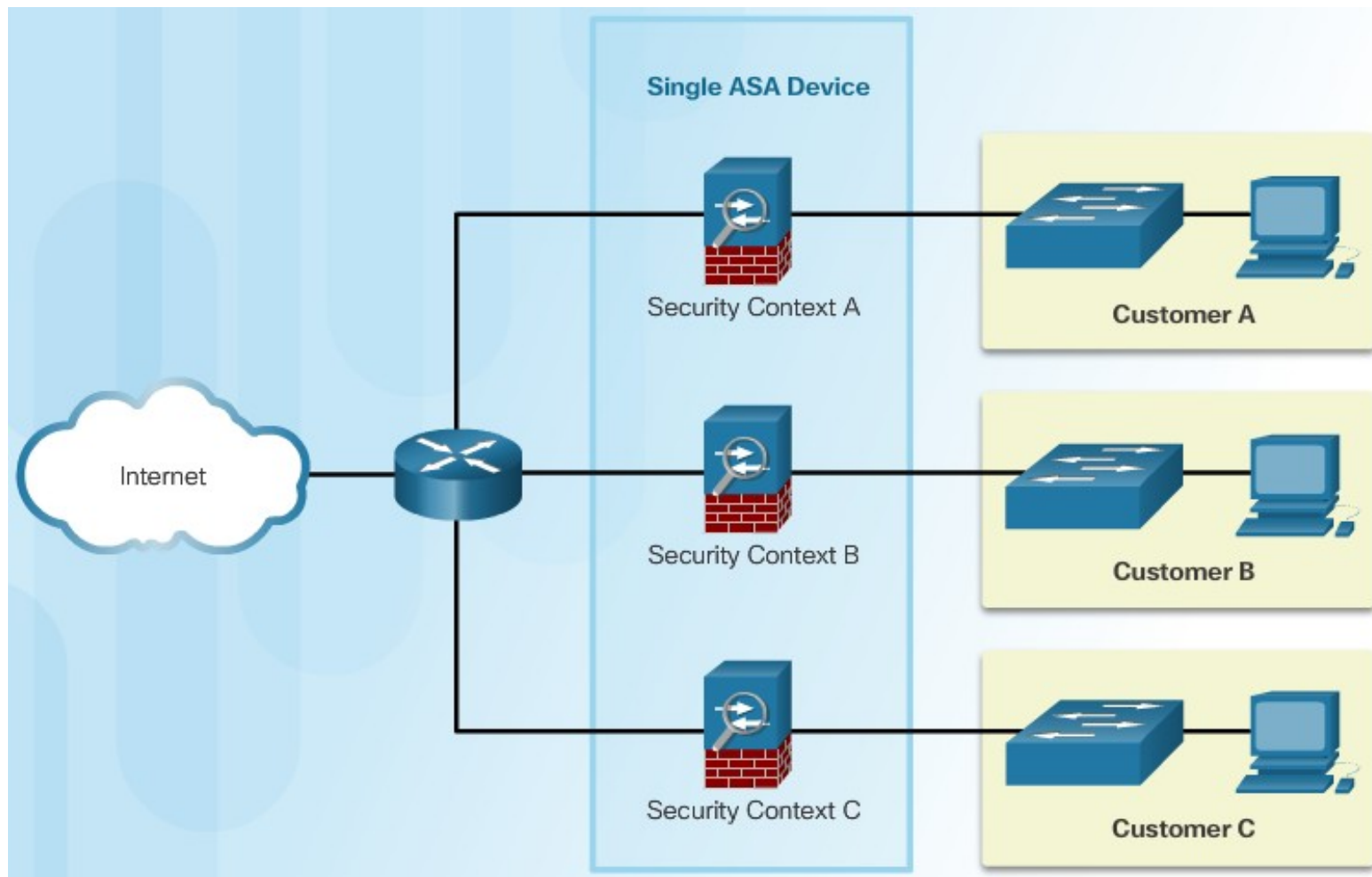


ASA Service Module

20 Gbps

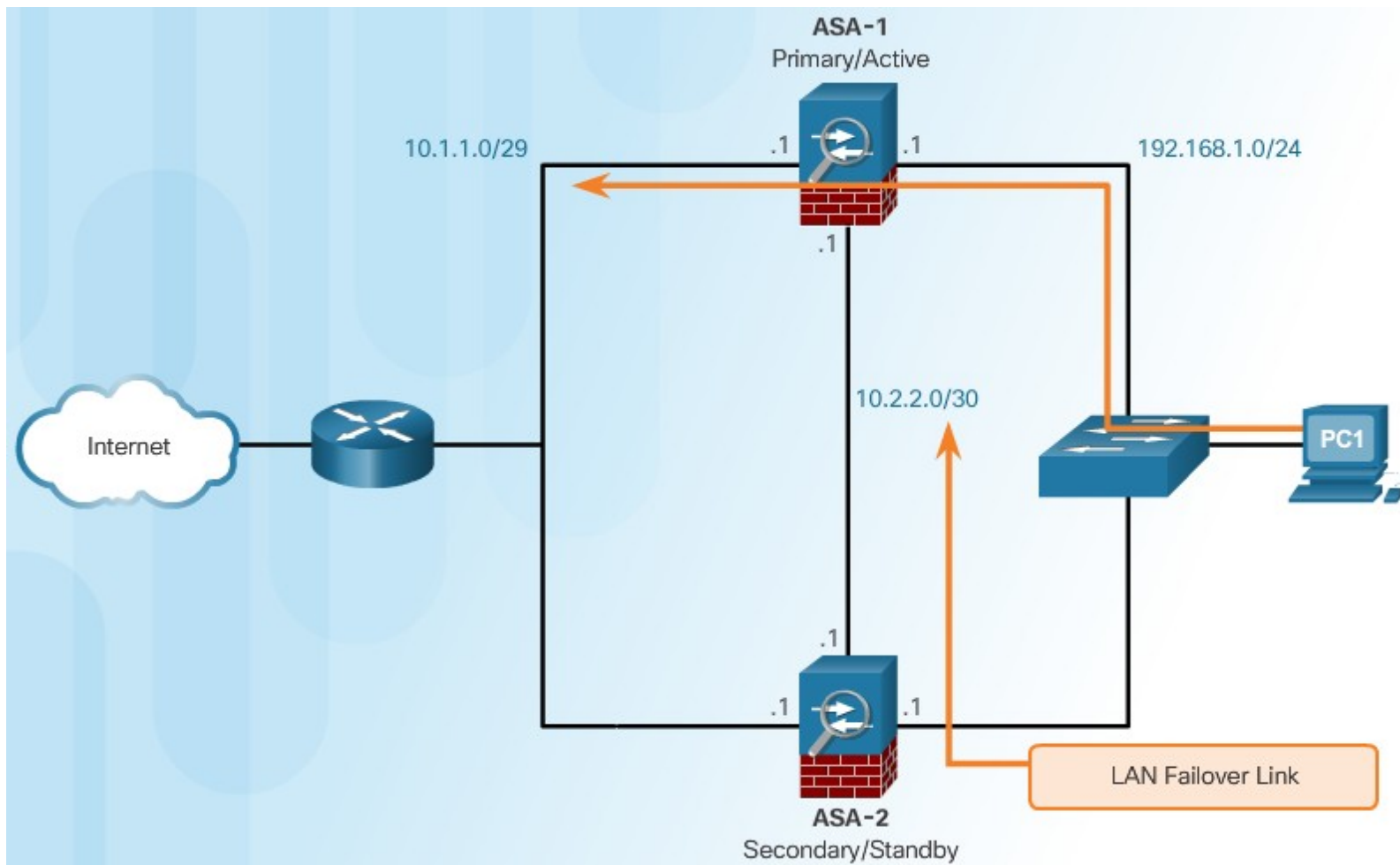
Kiterjesztett ASA tűzfal megoldások

ASA virtualizáció



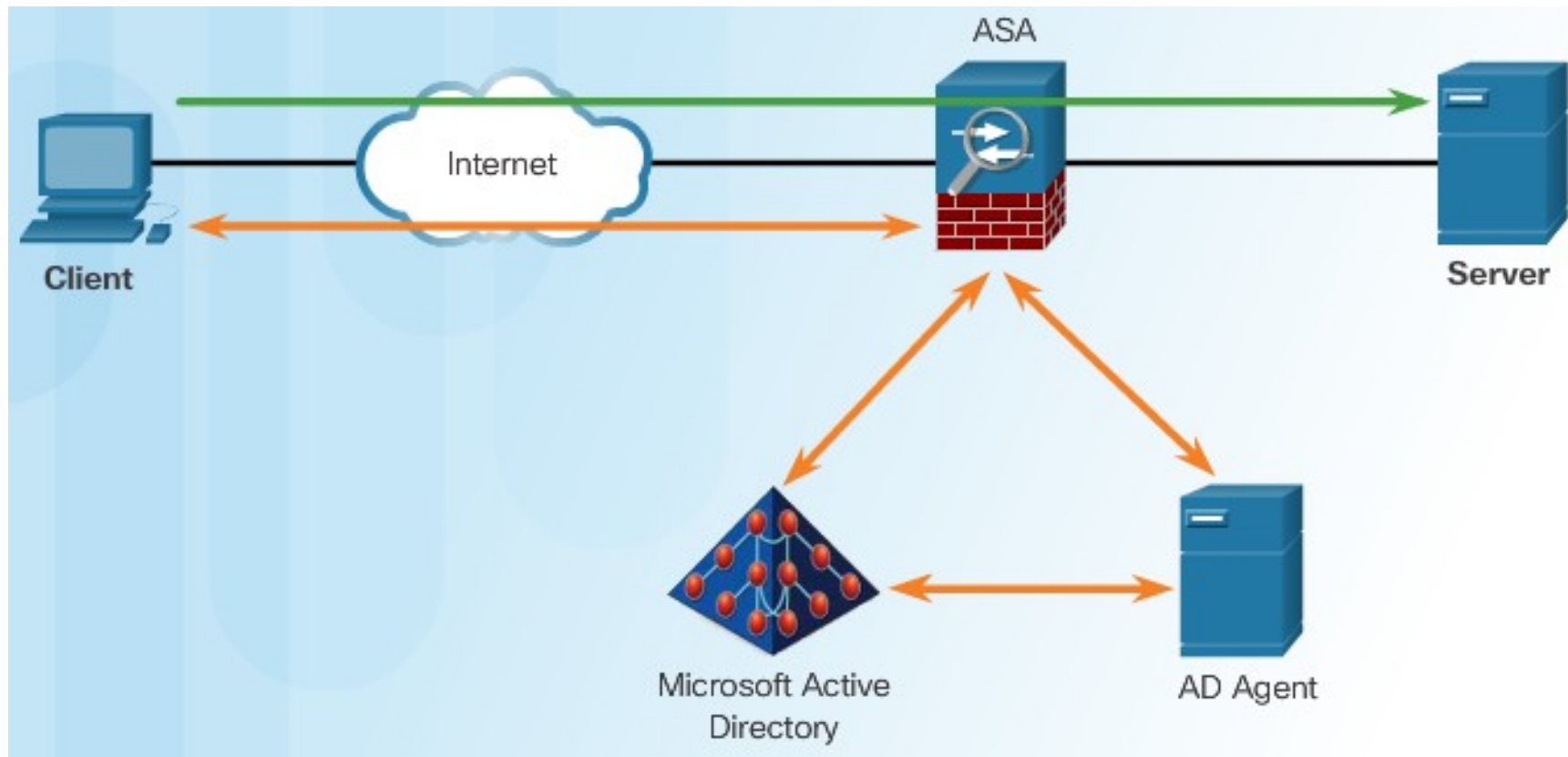
Kiterjesztett ASA tűzfal megoldások

Magas rendelkezésre állás



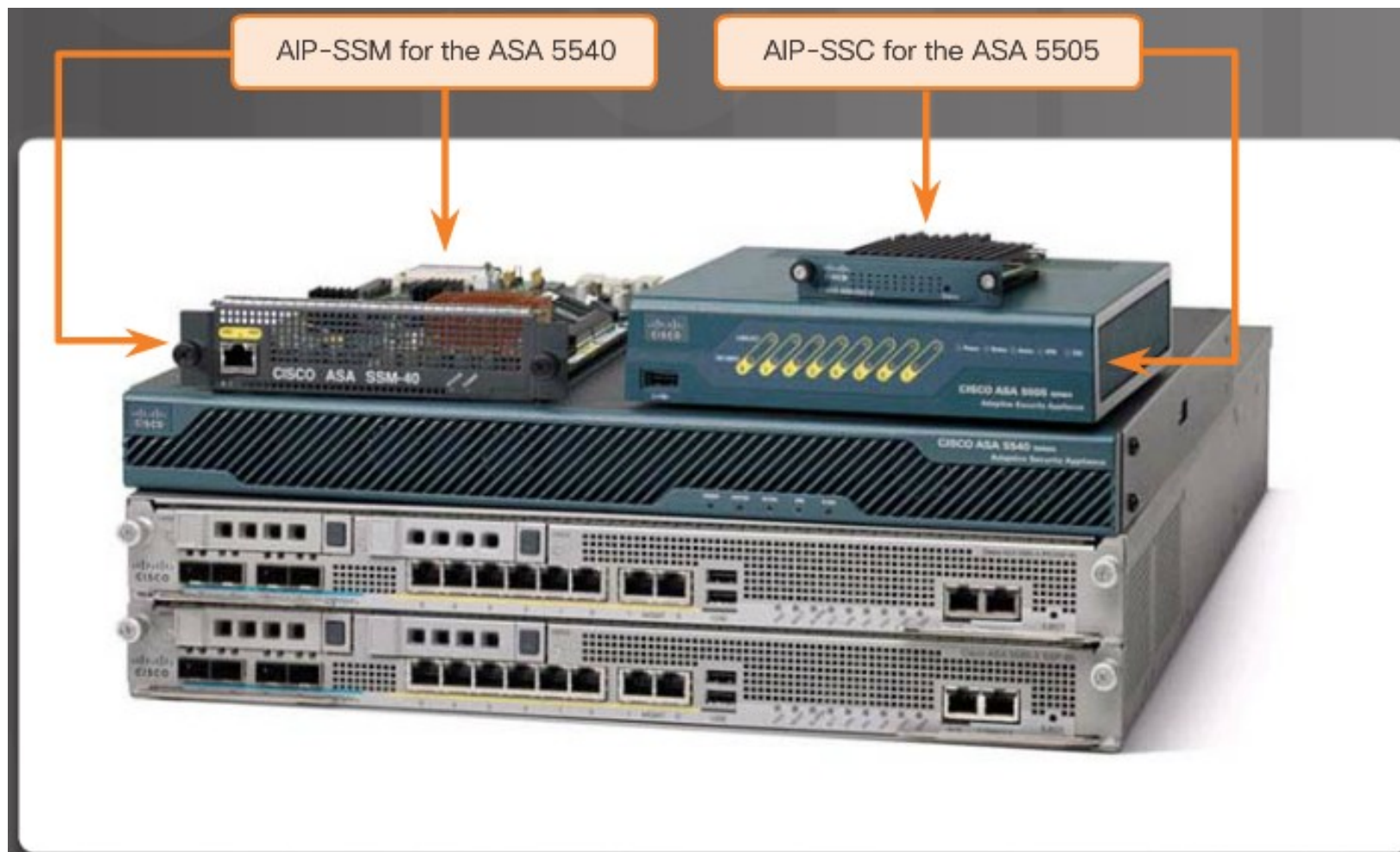
Kiterjesztett ASA tűzfal megoldások

Identity Firewall

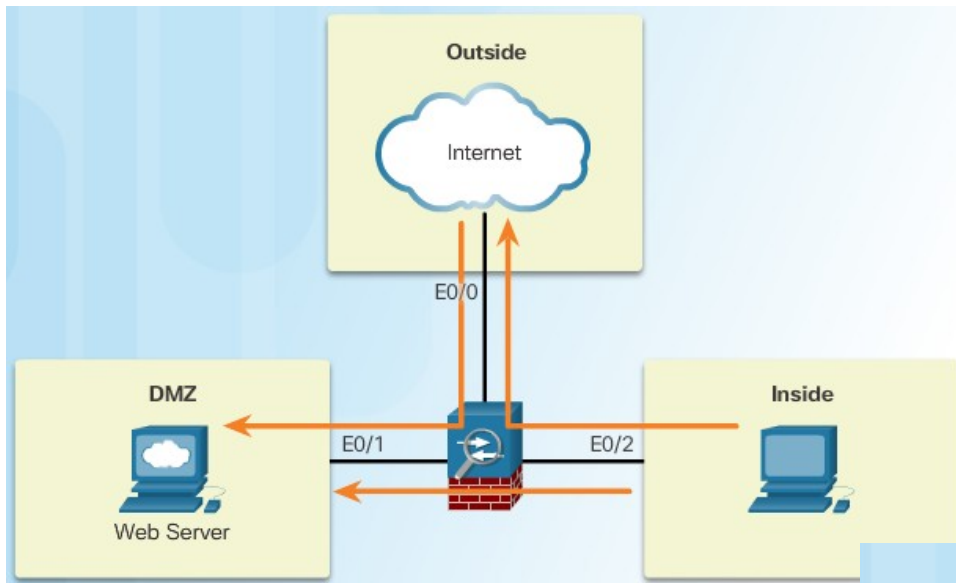


Kiterjesztett ASA tűzfal megoldások

ASA Threat Control

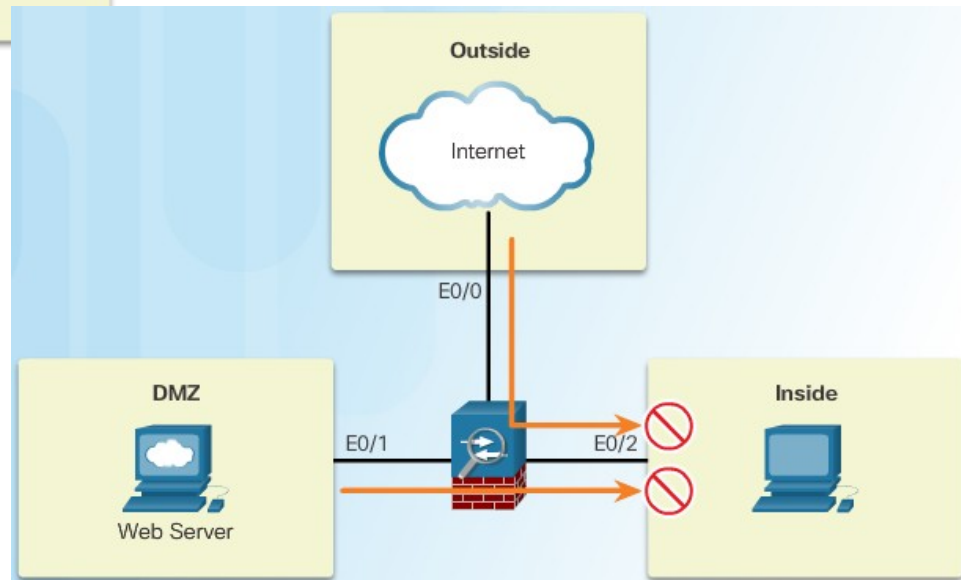


Tűzfal a hálózatban



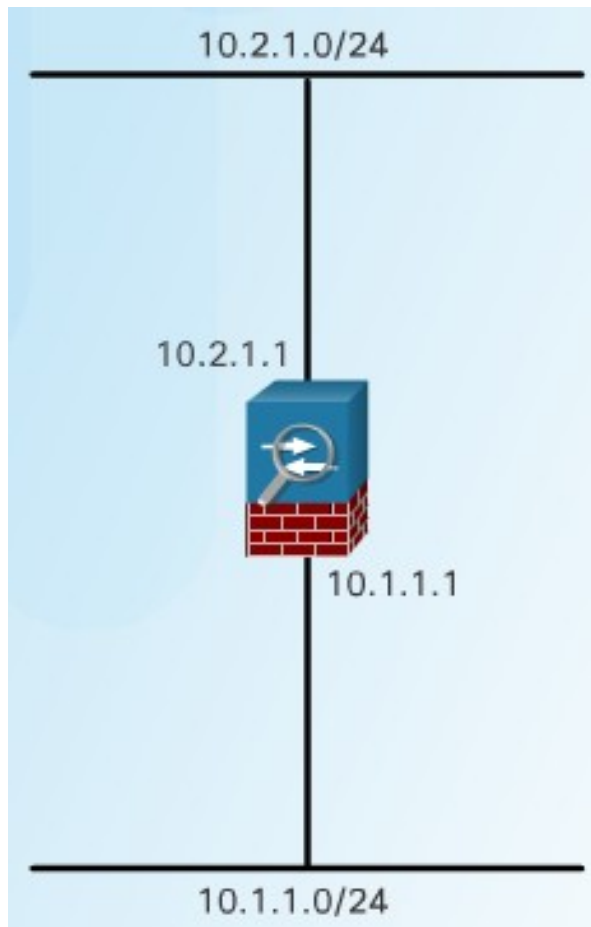
Engedélyezett
forgalom

Tiltott forgalom

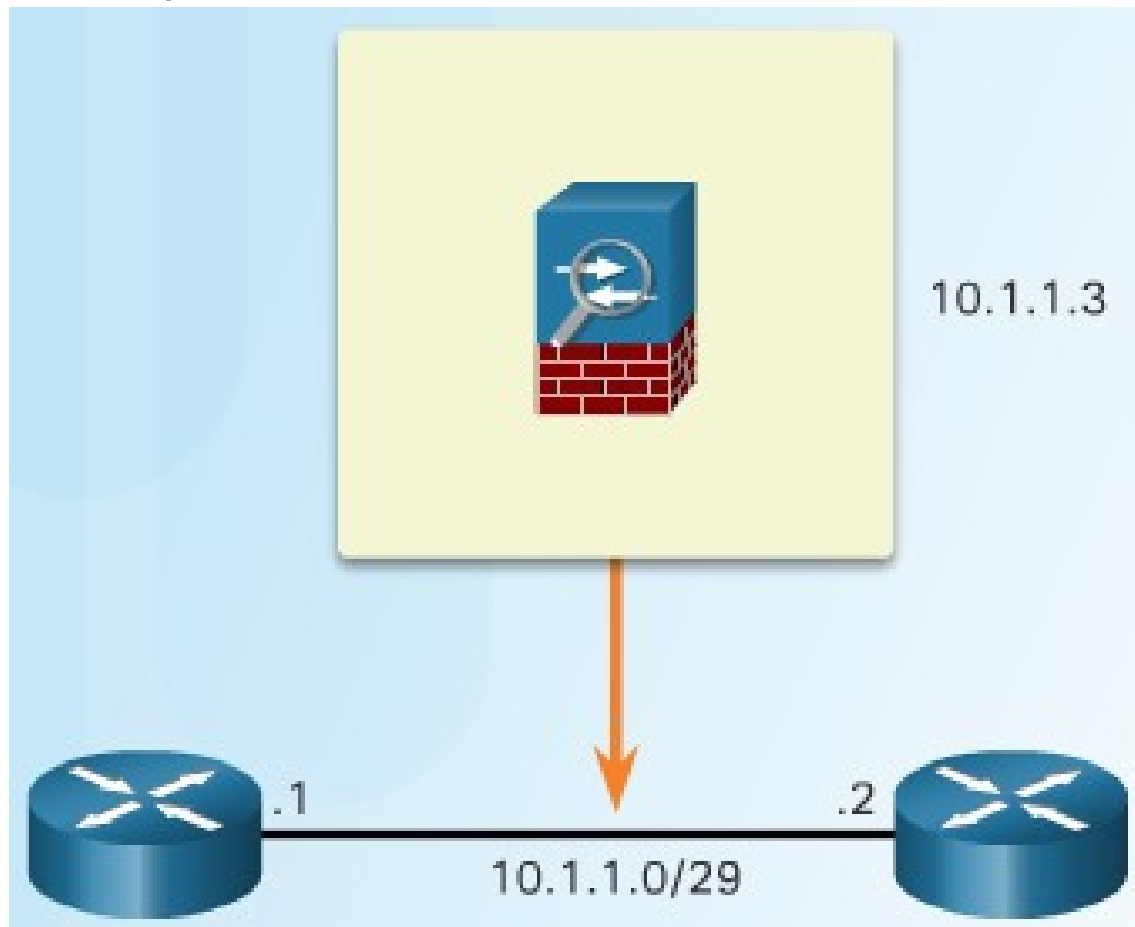


ASA tűzfal működési módok

Routed mód



Transparent mód



ASA licenszek

Base Licensz

Licenses		Description (Base License in Plaintext)		
Firewall Licenses				
Botnet Traffic Filter	Disabled	Optional Time-based license: Available		
Firewall Conns, Concurrent	10,000			
GTP/GPRS	No support			
Intercompany Media Engine	Disabled	Optional license: Available		
Unified Comm. Sessions	2	Optional license: 24		
VPN Licenses				
Adv. Endpoint Assessment	Disabled	Optional license: Available		
AnyConnect Essentials	Disabled	Optional license: Available (25 sessions)		
AnyConnect Mobile	Disabled	Optional license: Available		
AnyConnect Premium (sessions)	2	Optional Permanent or Time-based licenses:	10	25
Combined VPN sessions of all types, Maximum	25			
Other VPN (sessions)	10			
VPN Load Balancing	No Support			
VPN Licenses				
Encryption	Base (DES)	Opt. lic Strong (3DES/AES)		
Failover	Active/Standby (no stateful failover)			
Interfaces of all types, Max.	120			
Security Contexts	No Support			
Users, concurrent	10	Optional licenses:	50	Unlimited
VLANs/Zones, Maximum	Routed mode: 20			
	Transparent mode: 3 (2 regular zones and 1 failover link)			
VLAN Trunk, Maximum	8 trunks			

ASA licenszek

Licenses	Description (Security Plus Lic. in Plaintext)			
Firewall Licenses				
Botnet Traffic Filter	Disabled	Optional Time-based license: Available		
Firewall Conns, Concurrent	25,000			
GTP/GPRS	No support			
Intercompany Media Engine	Disabled	Optional license: Available		
Unified Comm. Sessions	2	Optional license: 24		
VPN Licenses				
Adv. Endpoint Assessment	Disabled	Optional license: Available		
AnyConnect Essentials	Disabled	Optional license: Available (25 sessions)		
AnyConnect Mobile	Disabled	Optional license: Available		
AnyConnect Premium (sessions)	2	Optional Permanent or Time-based licenses:	10	25
Combined VPN sessions of all types, Maximum	25			
Other VPN (sessions)	25			
VPN Load Balancing	No Support			
VPN Licenses				
Encryption	Base (DES)	Opt. lic Strong (3DES/AES)		
Failover	Active/Standby (no stateful failover)			
Interfaces of all types, Max.	120			
Security Contexts	No Support			
Users, concurrent	10	Optional licenses:	50	Unlimited
VLANs/Zones, Maximum	Routed mode: 20			
	Transparent mode: 3 (2 regular zones and 1 failover link)			
VLAN Trunk, Maximum	8 trunks			

Security Plus Licensz

ASA licenszek

show version paranccsal megtekinthetők a licensz információk

```
CCNAS-ASA# show version
<output omitted>

Licensed features for this platform:
Maximum Physical Interfaces      : 8           perpetual
VLANs                           : 3           DMZ Restricted
Dual ISPs                       : Disabled      perpetual
VLAN Trunk Ports                : 0           perpetual
Inside Hosts                    : 10          perpetual
Failover                        : Disabled      perpetual
Encryption-DES                  : Enabled      perpetual
Encryption-3DES-AES             : Enabled      perpetual
AnyConnect Premium Peers        : 2           perpetual
AnyConnect Essentials           : Disabled      perpetual
Other VPN Peers                 : 10          perpetual
Total VPN Peers                 : 12          perpetual
Shared License                   : Disabled      perpetual
AnyConnect for Mobile           : Disabled      perpetual
AnyConnect for Cisco VPN Phone  : Disabled      perpetual
Advanced Endpoint Assessment    : Disabled      perpetual
UC Phone Proxy Sessions         : 2           perpetual
Total UC Proxy Sessions         : 2           perpetual
Botnet Traffic Filter           : Disabled      perpetual
Intercompany Media Engine       : Disabled      perpetual
Cluster                         : Disabled      perpetual

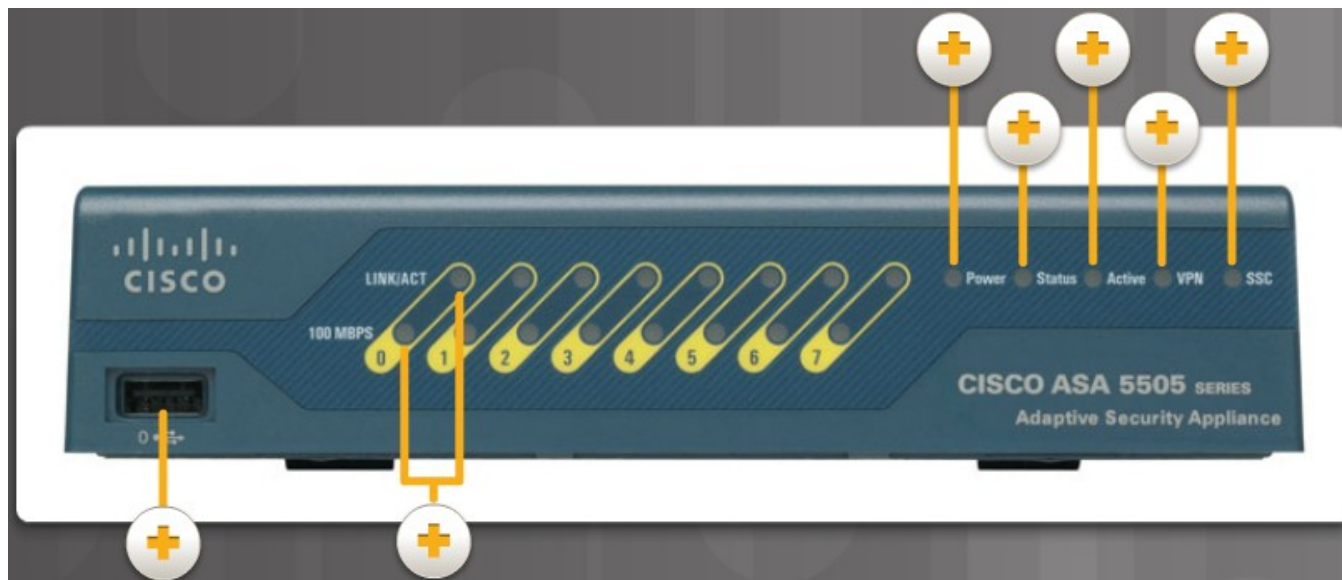
This platform has a Base license.
```

9.1.2

Az ASA beállítás alapjai

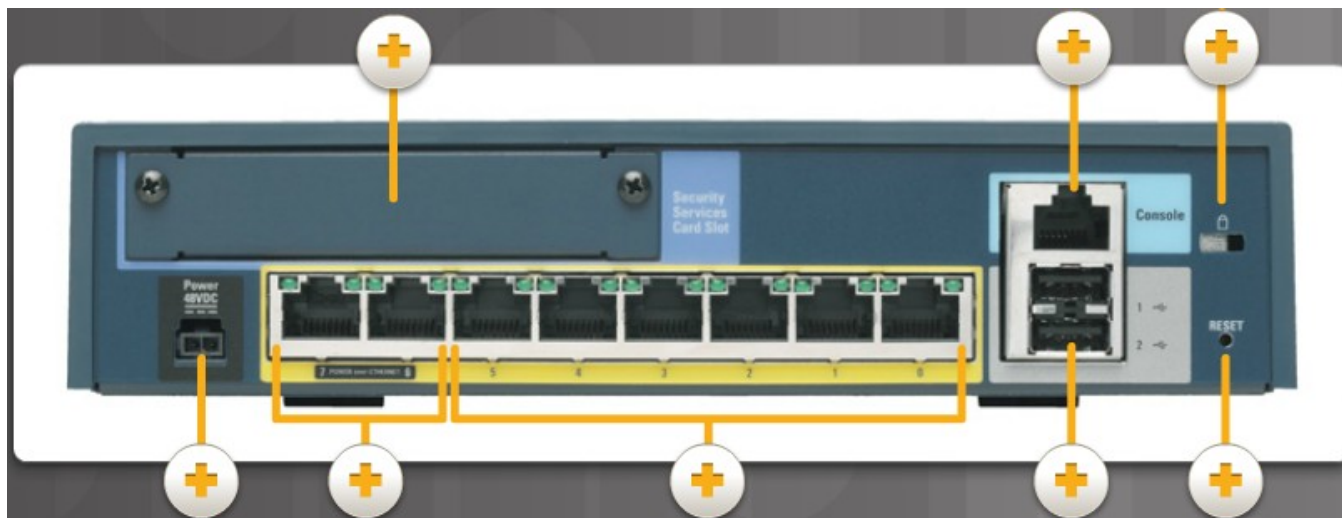


Az ASA 5505 áttekintése



ASA 5505 hátsó panel

ASA 5505 első panel



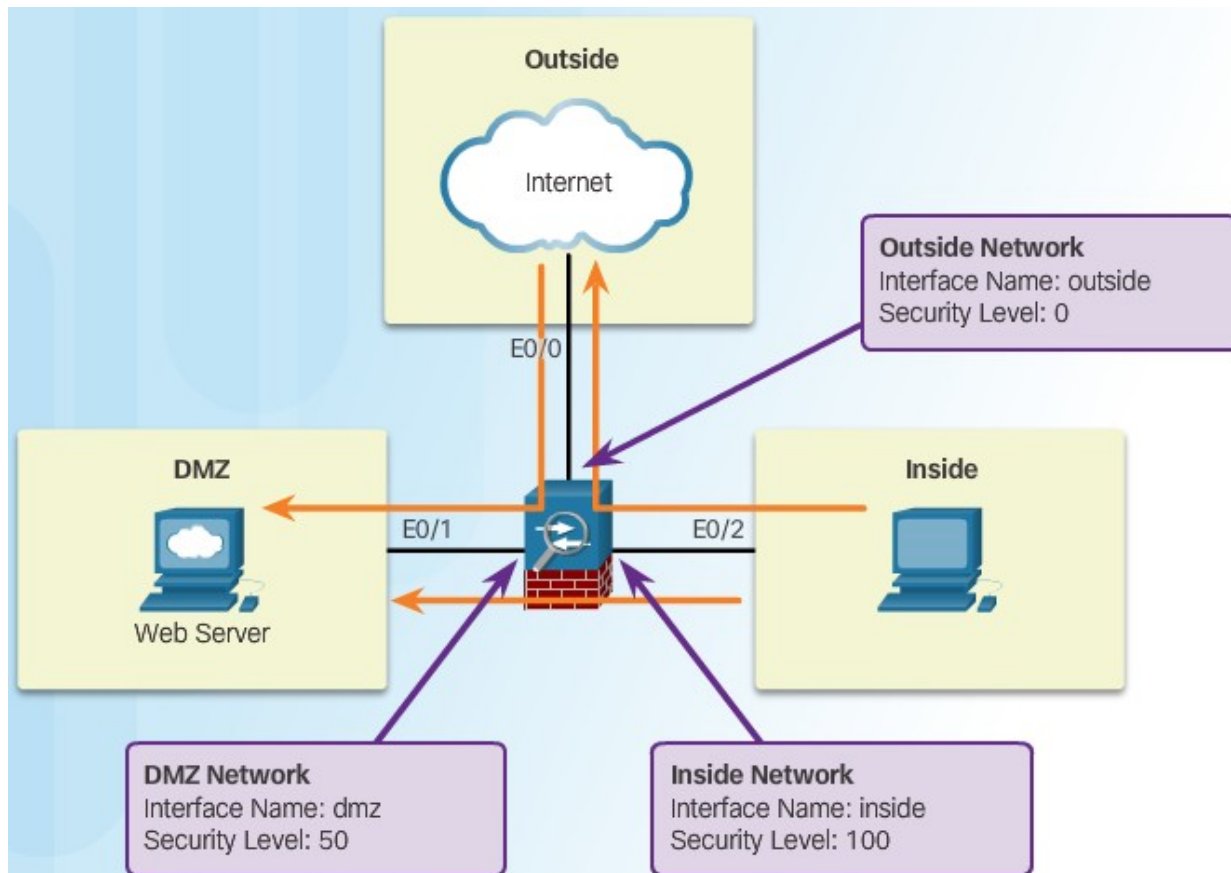
ASA biztonsági szintek

- Az ASA egy biztonsági szintet rendel minden interfészhez. Ezzel is különböztethetők meg az egyes zónák.
- Ez a biztonsági szint az interfész megbízhatóságát adja meg.
- Minél magasabb a szint az adott interfész annál megbízhatóbb.
- A biztonsági szint 0-100 közötti érték.
- Minden működő interfészhez nevet és biztonsági szintet kell rendelni.

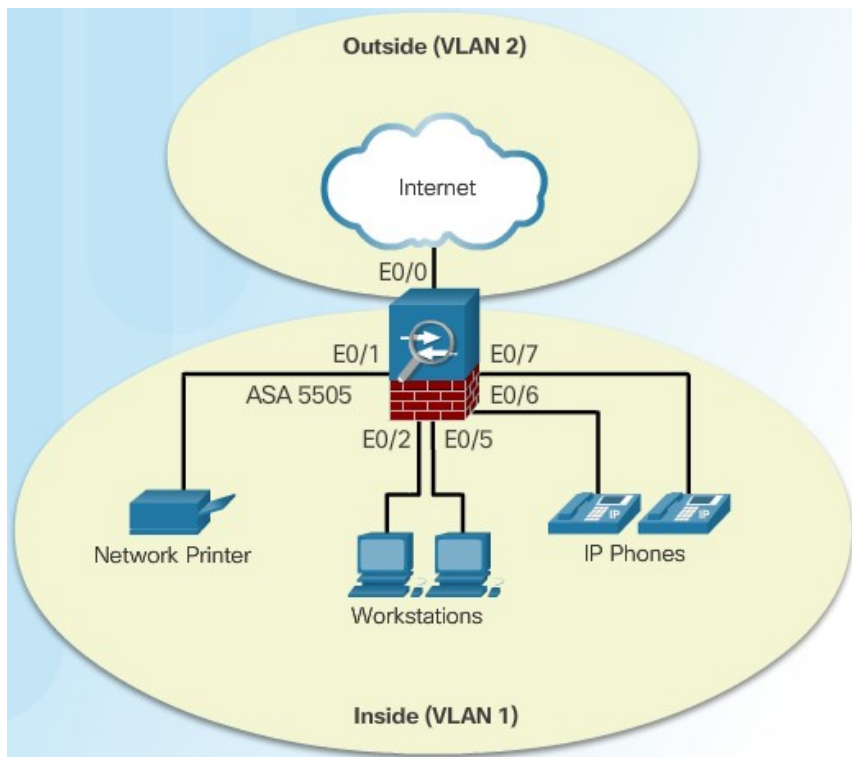
ASA biztonsági szintek

Biztonsági szint vezérlés:

- Network Access
- Inspection Engines
- Application Filtering

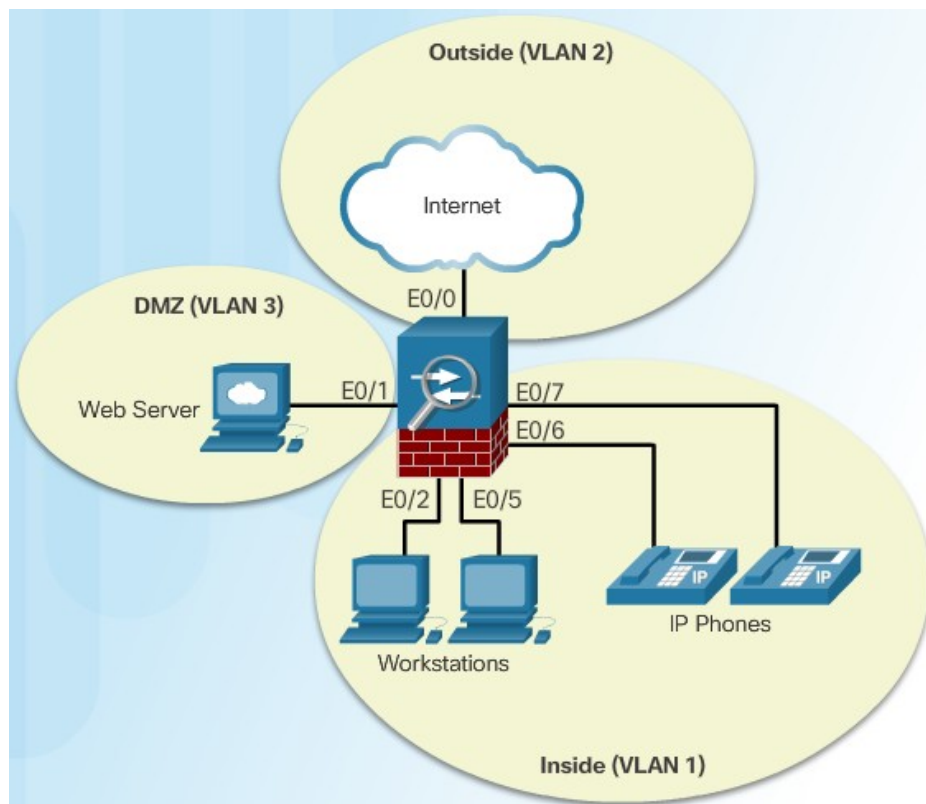


ASA 5505 megvalósítások



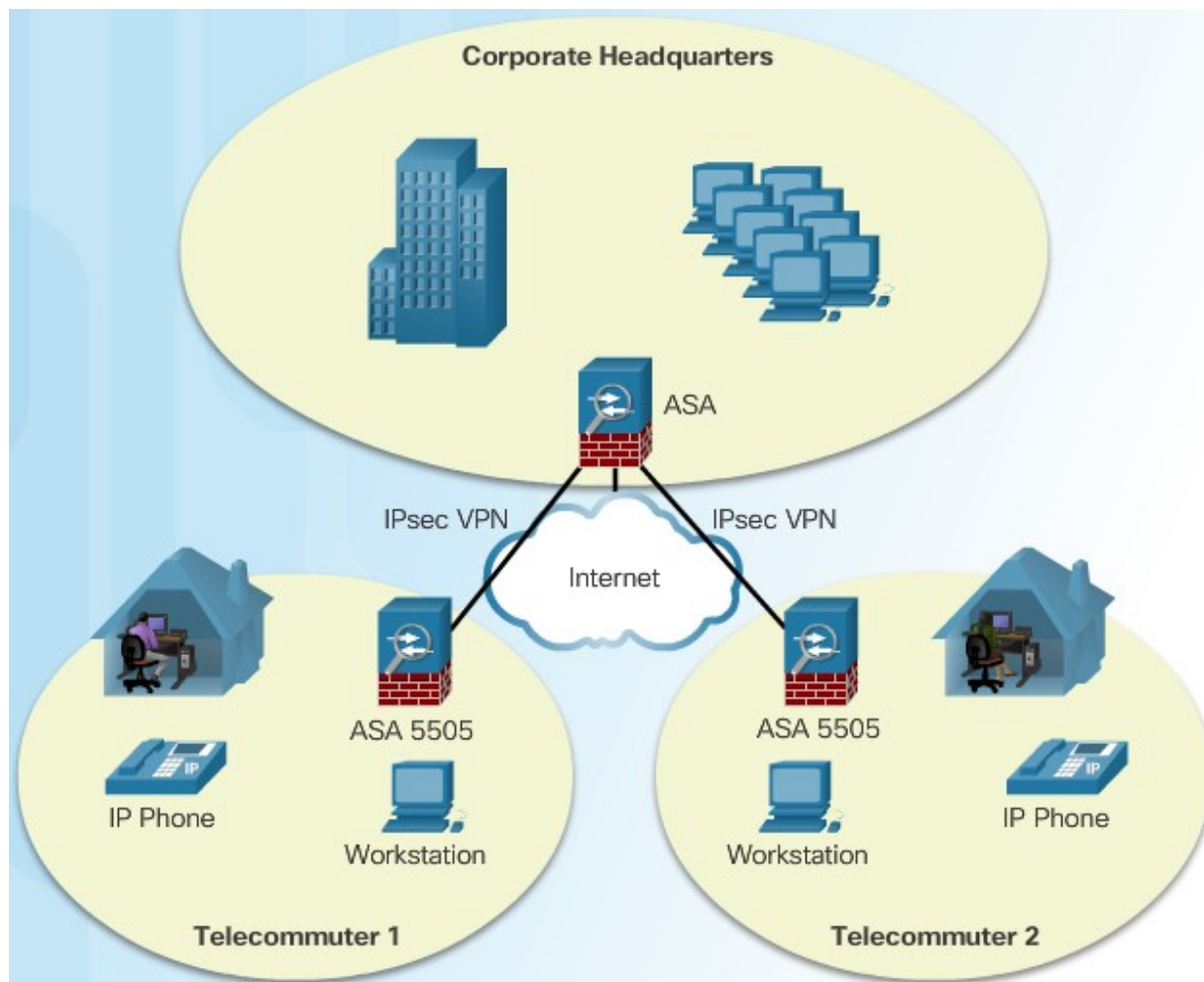
ASA megvalósítás egy
kisvállalkozásnál

ASA megvalósítás egy fiókirodánál



ASA 5505 megvalósítások

ASA megvalósítás nagyobb vállalatnál



9.2

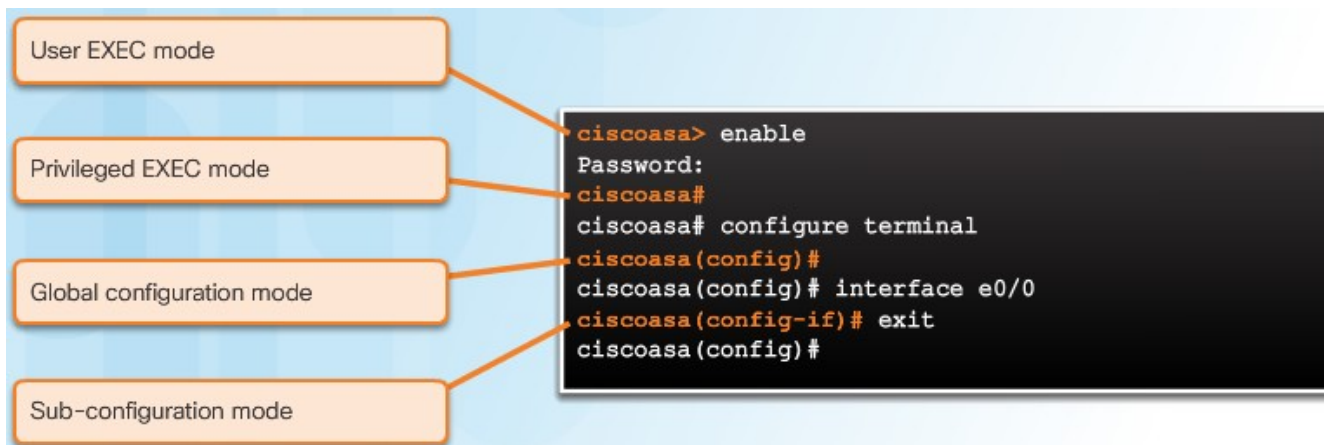
ASA tűzfal beállítása

9.2.1

Az ASA tűzfal beállítása



Alapszintű ASA beállítások



IOS vs. ASA parancsok

IOS Router Command	Equivalent ASA Command
enable secret <i>password</i>	enable password <i>password</i>
line vty 0 - 4 password <i>password</i> login	passwd <i>password</i>
ip route	route outside
show ip interfaces brief	show interfaces ip brief
show ip route	show route
show vlan	show switch vlan
show ip nat translations	show xlate
copy running-config startup-config	write [memory]
erase startup-config	write erase

Alapszintű ASA beállítások

show version parancs kimenete

```
ciscoasa# conf t
ciscoasa(config)# show password encryption
Password Encryption: Disabled
Master key hash: Not set(saved)
ciscoasa(config)#
ciscoasa(config)# help write
```

ASA CLI commands can be executed regardless of the current configuration mode prompt. The IOS **do** command is not required or recognized.

USAGE:

```
write erase|terminal|standby
write net [<tftp_ip>]:<filename>
write [memory]
```

The ASA provides a **help** command that provides a brief command description and syntax for certain commands.

DESCRIPTION:

```
write          Write config to net, flash, or terminal, or erase flash.
                Write without argument defaults to write memory
```

SYNTAX:

```
erase          Clears the flash memory configuration
```

```
terminal       Display the current active configuration, not necessarily
                the saved configuration
```

```
mem           Save the active configuration to the flash, so that it will
                be the active configuration after a reload
```

```
standby       Save the active configuration on the active unit to the
                flash on the standby unit
```

```
<--- More --->
```

To interrupt **show** command output, press the letter **Q**. The IOS **Ctrl+C (^C)** does not work.

ASA alapértelmezett konfiguráció

ASA 5505
alapértelmezett
beállítások

```
hostname ciscoasa
enable password 8Ry2YjIyt7RRXU24 encrypted names
!
interface Ethernet0/0
  switchport access vlan 2
!
interface Ethernet0/1
!
<output omitted>
interface Vlan1
  nameif inside
  security-level 100
  ip address 192.168.1.1 255.255.255.0
!
interface Vlan2
  nameif outside
  security-level 0
  ip address dhcp setroute
<output omitted>
object network obj_any
  nat (inside,outside) dynamic interface
<output omitted>
http server enable
http 192.168.1.0 255.255.255.0 inside
<output omitted>
dhcpd auto_config outside
!
dhcpd address 192.168.1.5-192.168.1.36 inside
dhcpd enable inside
<output omitted>
```

ASA interaktív beállítás varázsló

Belépés az ASA 5505 interaktív beállítás varázslóba

```
Pre-configure Firewall now through interactive prompts [yes]?
Firewall Mode [Routed]:
Enable password [<use current password>]: cisco
Allow password recovery [yes]?
Clock (UTC):
  Year [2015]:
  Month [Mar]: April
  Day [29]: 1
  Time [18:06:03]: 12:00:00
Management IP address: 192.168.1.1
Management network mask: 255.255.255.0
Host name: CCNAS-ASA
Domain name: ccnasecurity.com
IP address of host running Device Manager: 192.168.1.2

The following configuration will be used:
Enable password: cisco
Allow password recovery: yes
Clock (UTC): 12:00:00 April 1 2015
Firewall Mode: Routed
Management IP address: 192.168.1.1
Management network mask: 255.255.255.0
Host name: CCNAS-ASA
Domain name: ccnasecurity.com
IP address of host running Device Manager: 192.168.1.2

Use this configuration and save to flash? [yes]yes
INFO: Security level for "management" set to 0 by default.
```

9.2.2

Menedzsment beállítások és szolgáltatások konfigurálása



Belépés globális konfigurációs módba

Példa:

```
ciscoasa> enable
Password:
ciscoasa#
ciscoasa# clock set 12:00:00 1 April 2015
ciscoasa#
ciscoasa# configure terminal
ciscoasa(config)#

***** NOTICE *****

Help to improve the ASA platform by enabling anonymous reporting,
which allows Cisco to securely receive minimal error and health
information from the device. To learn more about this feature,
please visit: http://www.cisco.com/go/smartcall

Would you like to enable anonymous error reporting to help improve
the product? [Y]es, [N]o, [A]sk later: A
You will be reminded again in 7 days.

If you would like to enable this feature, issue the command
"call-home reporting anonymous".

Please remember to save your configuration.

ciscoasa(config)#
```

Alapbeállítások konfigurálása

Parancsok:

ASA Command	Description
hostname <i>name</i>	- Specifies a hostname up to 63 characters. - A hostname must start and end with a letter or digit, and have as interior characters only letters, digits, or a hyphen.
domain-name <i>name</i>	Sets the default domain name
enable password <i>password</i>	- Sets the enable password for privileged EXEC mode. - Sets the password as a case-sensitive string of 3 to 32 alphanumeric and special characters (not including a question mark or a space).
banner motd <i>message</i>	Provides legal notification and configures the system to display a message-of-the-day banner when connecting to the ASA
key config-key password-encryption [<i>new-pass</i> [<i>old-pass</i>]]	- Sets the passphrase between 8 and 128 character long. - Used for generation the encryption key.
password encryption aes	Enables password encryption and encrypts all user passwords.

Alapbeállítások konfigurálása

```
ciscoasa(config)# hostname CCNAS-ASA
CCNAS-ASA(config)# domain-name ccnasecurity.com
CCNAS-ASA(config)# enable password class
CCNAS-ASA(config)#
CCNAS-ASA(config)# banner motd -----
CCNAS-ASA(config)# banner motd     Authorized access only!
CCNAS-ASA(config)# banner motd     You have logged into a secure device.
CCNAS-ASA(config)# banner motd -----
CCNAS-ASA(config)# banner motd
CCNAS-ASA(config)# exit
CCNAS-ASA# exit
```

Logoff

```
-----
Authorized access only!
You have logged into a secure device.
-----
```

```
Type help or '?' for a list of available commands.
CCNAS-ASA>
```

Példa: AES titkosítás engedélyezése

```
CCNAS-ASA# show password encryption
Password Encryption: Disabled
Master key hash: Not set(saved)
CCNAS-ASA#
CCNAS-ASA# conf t
CCNAS-ASA(config)# key config-key password-encryption cisco123
CCNAS-ASA(config)# password encryption aes
CCNAS-ASA(config)# exit
CCNAS-ASA#
CCNAS-ASA# show password encryption
Password Encryption: Enabled
Master key hash: 0x45ebef8e 0x77a0f287 0x90247f80 0x2a184246 0xe85cbcc4(not saved)
CCNAS-ASA#
CCNAS-ASA# write
Building configuration...
Cryptochecksum: 99934042 e6c6b12b 607a9920 89d8a181

2359 bytes copied in 1.340 secs (2359 bytes/sec)
[OK]
CCNAS-ASA#
```

Alapbeállítások

Logikai VLAN interfészek beállítása

ASA Command	Description
<code>interface vlan vlan-number</code>	Enters VLAN interface configuration mode.
<code>nameif if_name</code>	- Names the interface using a text string of up to 48 characters. - The name is not case-sensitive. - You can change the name by re-entering this command with a new value. - Do not enter the no form, because that command causes all commands that refer to that name to be deleted.
<code>security-level value</code>	Sets the security level, where number is an integer between 0 (lowest) and 100 (highest).

Parancsok

IP cím beállítás a VLAN interfészen

To Configure	ASA Command	Description
Manually	<code>ip address ip-address netmask</code>	Assigns an IP address to the interface.
Using DHCP	<code>ip address dhcp</code>	Used to have the interface request an IP address configuration from the upstream device.
	<code>ip address dhcp setroute</code>	Used to have the interface request and install a default route to the upstream device.
Using PPPoE	<code>ip address pppoe</code>	Interface configuration mode command that requests an IP address from the upstream device.
	<code>ip address pppoe setroute</code>	Same command but it also requests and installs a default route to the upstream device.

Logikai VLAN interfészek beállítása

Példa:

```
CCNAS-ASA(config)# interface vlan 1
CCNAS-ASA(config-if)# nameif inside
INFO: Security level for "inside" set to 100 by default.
CCNAS-ASA(config-if)# security-level 100
CCNAS-ASA(config-if)# ip address 192.168.1.1 255.255.255.0
CCNAS-ASA(config-if)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# interface vlan 2
CCNAS-ASA(config-if)# nameif outside
INFO: Security level for "outside" set to 0 by default.
CCNAS-ASA(config-if)# security-level 0
CCNAS-ASA(config-if)# ip address 209.165.200.226 255.255.255.248
CCNAS-ASA(config-if)# exit
CCNAS-ASA(config)#
```

2. rétegbeli portok hozzárendelése a VLAN-hoz

```
CCNAS-ASA(config)# interface e0/0
CCNAS-ASA(config-if)# switchport access vlan 2
CCNAS-ASA(config-if)# no shut
CCNAS-ASA(config-if)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config-if)# interface e0/1
CCNAS-ASA(config-if)# switchport access vlan 1
CCNAS-ASA(config-if)# no shut
CCNAS-ASA(config-if)# exit
CCNAS-ASA(config)#
```

Példa

Ellenőrzés

```
CCNAS-ASA# show switch vlan
```

VLAN	Name	Status	Ports
1	inside	up	Et0/1, Et0/2, Et0/3, Et0/4 Et0/5, Et0/6, Et0/7
2	outside	up	Et0/0

```
CCNAS-ASA#
```

2. rétegbeli portok hozzárendelése a VLAN-hoz

```
CCNAS-ASA# show interface ip brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	unassigned	YES	unset	up	up
Ethernet0/1	unassigned	YES	unset	up	up
Ethernet0/2	unassigned	YES	unset	up	up
Ethernet0/3	unassigned	YES	unset	up	up
Ethernet0/4	unassigned	YES	unset	down	down
Ethernet0/5	unassigned	YES	unset	down	down
Ethernet0/6	unassigned	YES	unset	down	down
Ethernet0/7	unassigned	YES	unset	down	down
Internal-Data0/0	unassigned	YES	unset	up	up
Internal-Data0/1	unassigned	YES	unset	up	up
Vlan1	192.168.1.1	YES	manual	up	up
Vlan2	209.165.200.226	YES	manual	up	up
Virtual0	127.1.0.1	YES	unset	up	up

```
CCNAS-ASA#
```

Ellenőrzés

IP címek
ellenőrzése

```
CCNAS-ASA# show ip address
```

System IP Addresses:

Interface	Name	IP address	Subnet mask	Method
Vlan1	inside	192.168.1.1	255.255.255.0	manual
Vlan2	outside	209.165.200.226	255.255.255.248	manual

Current IP Addresses:

Interface	Name	IP address	Subnet mask	Method
Vlan1	inside	192.168.1.1	255.255.255.0	manual
Vlan2	outside	209.165.200.226	255.255.255.248	manual

```
CCNAS-ASA#
```

Alapértelmezett statikus út definiálás

- Ha az ASA DHCP kliens, akkor az alapértelmezett útvonal adatait megkapja a másik eszköztől. Ha nem, akkor nekünk kell megadni az útvonalat.
- **route** *if_nev 0.0.0.0 0.0.0.0 kov_ugras_IP*

```
CCNAS-ASA(config)# route outside 0.0.0.0 0.0.0.0 209.165.200.225
CCNAS-ASA(config)#
CCNAS-ASA(config)# show route | begin Gateway
Gateway of last resort is 209.165.200.225 to network 0.0.0.0

S*      0.0.0.0 0.0.0.0 [1/0] via 209.165.200.225, outside
C        192.168.1.0 255.255.255.0 is directly connected, inside
L        192.168.1.1 255.255.255.255 is directly connected, inside
C        209.165.200.224 255.255.255.248 is directly connected, outside
L        209.165.200.226 255.255.255.255 is directly connected, outside

CCNAS-ASA(config)#
```

Távoli elérések beállítása - telnet

- **passwd jelszo** jelszó megadása
- **telnet ip maszk if_nev** IP címek ill. interfészek megadása, ahonnan betelnetelhet a kliens.
- **telnet timeout perc** alapértelmezett 5 perc
- **aaa authentication telnet console LOCAL**
telnet bejelentkezés a helyi adatbázis alapján. A passwd parancsot felülírja.
- **clear configure telnet** A telnet kapcsolatokat eltávolítja a konfigurációból.

```
CCNAS-ASA(config)# password cisco
CCNAS-ASA(config)# telnet 192.168.1.3 255.255.255.255 inside
CCNAS-ASA(config)# telnet timeout 3
CCNAS-ASA(config)#
CCNAS-ASA(config)# show run telnet
telnet 192.168.1.3 255.255.255.255 inside
telnet timeout 3
CCNAS-ASA(config)#
```

Távoli elérések beállítása - SSH

- **username** *nev* **password** *jelszo*
- **aaa authentication ssh console LOCAL**
- **crypto key generate rsa modulus** *szam*
 - A kulcshossz 512, 768, 1024 vagy 2048 lehet.
- **ssh ip maszk** *if_nev*
 - Meghatározza, hogy melyik belső interfészről és milyen IP című kliens ssh-zhat be.
- **ssh version** *szam*
 - Alapértelmezett 1 és 2.
- **ssh timeout** *perc*
 - Alapértelmezett 5 perc.

Távoli elérések beállítása - SSH

```
CCNAS-ASA(config)# username ADMIN password class
CCNAS-ASA(config)#
CCNAS-ASA(config)# aaa authentication ssh console LOCAL
CCNAS-ASA(config)#
CCNAS-ASA(config)# crypto key generate rsa modulus 2048
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.

Do you really want to replace them? [yes/no]: y
Keypair generation process begin. Please wait...
CCNAS-ASA(config)#
CCNAS-ASA(config)# ssh 192.168.1.3 255.255.255.255 inside
CCNAS-ASA(config)# ssh 192.168.1.4 255.255.255.255 inside
CCNAS-ASA(config)# ssh 172.16.1.3 255.255.255.255 outside
CCNAS-ASA(config)#
CCNAS-ASA(config)# ssh version 2
CCNAS-ASA(config)#
CCNAS-ASA(config)# show ssh
Timeout: 5 minutes
Version allowed: 2
192.168.1.3 255.255.255.255 inside
192.168.1.4 255.255.255.255 inside
172.16.1.3 255.255.255.255 outside
CCNAS-ASA(config)#
```

NTP beállítás

NTP hitelesítés beállítás parancsok

ASA Command	Description
<code>ntp authenticate</code>	• Enables authentication with an NTP server.
<code>ntp trusted-key <i>key_id</i></code>	• Specifies an authentication key ID to be a trusted key, which is required for authentication with an NTP server.
<code>ntp authentication-key <i>key_id</i> md5 <i>key</i></code>	• Sets a key to authenticate with an NTP server.
<code>ntp server <i>ip_address</i> [<i>key key_id</i>]</code>	• Identifies an NTP server.

Példa:

```
CCNAS-ASA(config)# ntp authenticate
CCNAS-ASA(config)# ntp trusted-key 1
CCNAS-ASA(config)# ntp authentication-key 1 md5 cisco123
CCNAS-ASA(config)# ntp server 192.168.1.254
CCNAS-ASA(config)#
```

DHCP szerver beállítások

- **dhcpcd address** *kezdolP – utolsoIP if_nev*
 - Basic licensszel 32 kliens a maximum. Habár a Basic licensz esetén 10 klienst enged meg egyidőben csatlakozni.
- **dhcpcd dns** *dns_cim*
- **dhcpcd lease** *masodperc*
 - Alapértelmezett 3600s (1 óra)
- **dhcpcd domain** *domain_nev*
- **dhcpcd enable** *if_nev*
- Show parancsok:
 - **show dhcpcd state, show dhcpcd binding, show dhcpcd statistics**

DHCP szerver beállítások

```
CCNAS-ASA(config)# dhcpd address 192.168.1.10-192.168.1.100
ERROR: % Incomplete command
CCNAS-ASA(config)# dhcpd address 192.168.1.10-192.168.1.100 inside
Warning, DHCP pool range is limited to 32 addresses, set address range as:
192.168.1.10-192.168.1.41
CCNAS-ASA(config)# dhcpd address 192.168.1.10-192.168.1.41 inside
CCNAS-ASA(config)# dhcpd lease 1800
CCNAS-ASA(config)#
```

9.2.3

Objektum csoportok



Objektumok és objektum csoportok

- Objektum: Az ASA által létrehozott és használt gyűjtőnév a belső IP-kre, bármilyen konfigurációra. Lehet hoszt, hálózati IP cím, IP tartomány, protokoll, port tartomány.
- Előnye, hogy többszörösen felhasználható.
- 1 objektum 1 vagy több objektum csoporthoz is tartozhat.
- Két módon hozható létre objektum:
 - Hálózati objektum (network object)
 - Szolgáltatás objektum (service object)

Objektumok és objektum csoportok

```
CCNAS-ASA(config)# object ?
```

```
configure mode commands/options:
```

```
network    Specifies a host, subnet or range IP addresses
```

```
service    Specifies a protocol/port
```

```
CCNAS-ASA(config)#
```

```
CCNAS-ASA(config)# object-group ?
```

```
configure mode commands/options:
```

```
icmp-type  Specifies a group of ICMP types, such as echo
```

```
network    Specifies a group of host or subnet IP addresses
```

```
protocol   Specifies a group of protocols, such as TCP, etc
```

```
service    Specifies a group of TCP/UDP ports/services
```

```
user       Specifies single user, local or import user group
```

```
CCNAS-ASA(config)#
```

Hálózati objektumok létrehozása

Object network *nev* paranccsal

ASA Command	Description
host <i>ip-addr</i>	• Assigns an IP address to the named object.
subnet <i>net-address net-mask</i>	• Assigns a network subnet to the named object.
range <i>ip-addr-1 ip-addr-n</i>	• Assigns IP addresses in a range

Configuring a Network Object Example

```
CCNAS-ASA(config)# object network EXAMPLE-1
CCNAS-ASA(config-network-object)# host 192.168.1.3
CCNAS-ASA(config-network-object)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# show running-config object
object network EXAMPLE-1
  host 192.168.1.3
CCNAS-ASA(config)#
CCNAS-ASA(config)# object network EXAMPLE-1
CCNAS-ASA(config-network-object)# host 192.168.1.4
CCNAS-ASA(config-network-object)# range 192.168.1.10 192.168.1.20
CCNAS-ASA(config-network-object)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# show running-config object
object network EXAMPLE-1
  range 192.168.1.10 192.168.1.20
CCNAS-ASA(config)#
```

Szolgáltatás objektum létrehozása

- **object service** *nev* paranccsal lehet létrehozni.
- Protokollt, forrás és cél portot tartalmazhat.

ASA Command

```
service protocol [ source [ operator  
port ]] [ destination [ operator port]]
```

```
service tcp [ source [ operator port ]]  
[ destination [ operator port ]]
```

```
service udp [ source [ operator port ]]  
[ destination [ operator port ]]
```

```
service icmp icmp-type
```

```
service icmp6 icmp6-type
```

Szolgáltatás objektumok létrehozása

Példa:

```
CCNAS-ASA(config)# object service EXAMPLE-2
CCNAS-ASA(config-service-object)#
CCNAS-ASA(config-service-object)# service ?

service-object mode commands/options:
  <0-255> Enter protocol number (0 - 255)
  ah
  eigrp
  esp
  gre
  icmp
  icmp6
  igmp
  igrp
  ip
  ipinip
  ipsec
  nos
  ospf
  pcp
  pim
  pptp
  snp
  tcp
  udp

configure mode commands/options:
  call-home          Enable or disable Smart Call-Home
  internal           Advanced settings (see below under Class command)
```

Szolgáltatás objektumok létrehozása

Gyakori szolgáltatás objektum parancsok

ASA Command	Description
service <i>protocol</i> [source [<i>operator port</i>]] [destination [<i>operator port</i>]]	Specifies an IP protocol name or number.
service tcp [source [<i>operator port</i>]] [destination [<i>operator port</i>]]	Specifies that the service object is for the TCP protocol.
service udp [source [<i>operator port</i>]] [destination [<i>operator port</i>]]	Specifies that the service object is for the UDP protocol.
service icmp <i>icmp-type</i>	Specifies that the service object is for the ICMP protocol.
service icmp6 <i>icmp6-type</i>	Specifies that the service object is for the ICMPv6 protocol.

Példa:

```
CCNAS-ASA(config)# object service SERV-1
CCNAS-ASA(config-service-object)# service tcp destination eq ftp
CCNAS-ASA(config-service-object)# service tcp destination eq www
CCNAS-ASA(config-service-object)# exit
CCNAS-ASA(config)# show running-config object service
object service SERV-1
  service tcp destination eq www
CCNAS-ASA(config)#
```

Objektum csoportok

- Az objektum csoport az objektumok csoportosításával jön létre.
- Az objektum csoportot egy ACE-ként (Access Control Entry) lehet használni.
- Az ASA a következő objektum csoportokat támogatja:
 - Hálózati (Network)
 - Szolgáltatás (Service)
 - Biztonsági (Security) Cisco TrustSec támogatására
 - ICMP típusú (ICMP-type)
 - Felhasználói (User) Helyi és importált AD

Objektum csoportok konfigurálása

object-group network *group_nev*

network-object (Itt a már megírt objektum nevét adhatjuk meg vagy megírjuk magát az objektumot.)

group-object *grp_nev*

object-group icmp-type *group_nev*

icmp-object *icmp_típus*

group-object *grp_nev*

object-group service *group_nev*

service-object *obj_nev*

group-object *grp_nev*

port-object eq *port* | **range** *kezd_port veg_port*

Gyakori objektum csoportok beállítása

```
CCNAS-ASA(config)# object-group network ADMIN-HOST
CCNAS-ASA(config-network-object-group)# description Administrative hosts
CCNAS-ASA(config-network-object-group)# network-object host 192.168.1.3
CCNAS-ASA(config-network-object-group)# network-object host 192.168.1.4
CCNAS-ASA(config-network-object-group)# exit
CCNAS-ASA(config)# object-group network ALL-HOSTS
CCNAS-ASA(config-network-object-group)# description All inside hosts
CCNAS-ASA(config-network-object-group)# network-object 192.168.1.32 255.255.255.240
CCNAS-ASA(config-network-object-group)# group-object ADMIN-HOST
CCNAS-ASA(config-network-object-group)# exit
CCNAS-ASA(config)# show run object-group
object-group network ADMIN-HOST
  description Administrative host IP addresses
  network-object host 192.168.1.3
  network-object host 192.168.1.4
object-group network ALL-HOSTS
  network-object 192.168.1.32 255.255.255.240
  group-object ADMIN-HOST
CCNAS-ASA(config)#
```

Példa a hálózati objektum csoport beállítására

Példa ICMP-típusú objektum csoport beállítására

```
CCNAS-ASA(config)# object-group icmp-type ICMP-ALLOWED
CCNAS-ASA(config-icmp-object-group)# icmp-object echo
CCNAS-ASA(config-icmp-object-group)# icmp-object time-exceeded
CCNAS-ASA(config-icmp-object-group)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# show running-config object-group id ICMP-ALLOWED
object-group icmp-type ICMP-ALLOWED
  icmp-object echo
  icmp-object time-exceeded
CCNAS-ASA(config)#
```

Gyakori objektum csoportok beállítása

Példa szolgáltatás objektum csoport beállítására

```
CCNAS-ASA(config)# object-group service SERVICES-1
CCNAS-ASA(config-service-object-group)# service-object tcp destination eq www
CCNAS-ASA(config-service-object-group)# service-object tcp destination eq https
CCNAS-ASA(config-service-object-group)# service-object tcp destination eq pop3
CCNAS-ASA(config-service-object-group)# service-object udp destination eq ntp
CCNAS-ASA(config-service-object-group)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# object-group service SERVICES-2 tcp
CCNAS-ASA(config-service-object-group)# port-object eq www
CCNAS-ASA(config-service-object-group)# port-object eq smtp
CCNAS-ASA(config-service-object-group)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# object-group service SERVICES-3 tcp
CCNAS-ASA(config-service-object-group)# group-object SERVICES-2
CCNAS-ASA(config-service-object-group)# port-object eq ftp
CCNAS-ASA(config-service-object-group)# port-object range 2000 2005
CCNAS-ASA(config-service-object-group)# exit
CCNAS-ASA(config)#
```

9.2.4 ACL-ek



ASA ACL-ek

- Hasonló az IOS ACL-ekhez.
- Különbség:
 - A hálózati maszkot használ a wildcard maszk helyett.
 - Nevesített ACL-eket használ.
 - Ha nincs ACL a biztonsági szint akkor is szűr.

Az ACL szűrők típusai

- Az ASA nem csak azt a forgalmat tudja szűrni, amely áthalad az eszközön, hanem azt is, aminek a célja az ASA.
- Kétféle szűrési típus van.
 - Through-traffic filtering – Azt a forgalmat szűri, amely egyik interfészről áthalad a másakra. Két lépésben hozható létre.
 - To-the-box-traffic filtering – (management access role)
Az ASA-hoz érkező forgalom szűrésére használható. Egy lépéssel konfigurálható, de kiegészítő szabályok kellenek az access control részen.

Az ACL szűrők típusai

- A biztonsági szintek ACL nélkül is szűrik a forgalmat.
- A nagyobb biztonsági szint felől a kisebb felé engedélyezett a forgalom fordítva nem.
- A biztonsági szinttel szabályozott hozzáférés felülírható ACL-lel.
- Az azonos biztonsági szintű interfészek közötti kommunikációhoz adjuk ki a **same-security-traffic permit inter-interface** parancsot.
- Ha azt is engedélyezni akarjuk, hogy a bejövő interfészen ki is menjen a forgalom, akkor a **same-security-traffic permit intra-interface** parancsot kell kiadni.

ASA ACL típusok

- 5 típust támogat
 - Kiterjesztett ACL – Ezt használják hálózati hozzáférés vezérlésénél, AAA-nál, NAT-nál, VPN-nél, MPF esetén.
 - Standard ACL – Tipikusan OSPF útvonalaknál használják.
 - Ether-type ACL – Csak ha transparent módban van az eszköz.
 - Web-type ACL – A clientless SSL VPN szűrésére.
 - IPv6 ACL

ACL-ek konfigurálása

help access-list parancs

□ access-list parancs paraméterei

```
CCNAS-ASA(config)# help access-list
```

```
USAGE:
```

```
Extended access list:
```

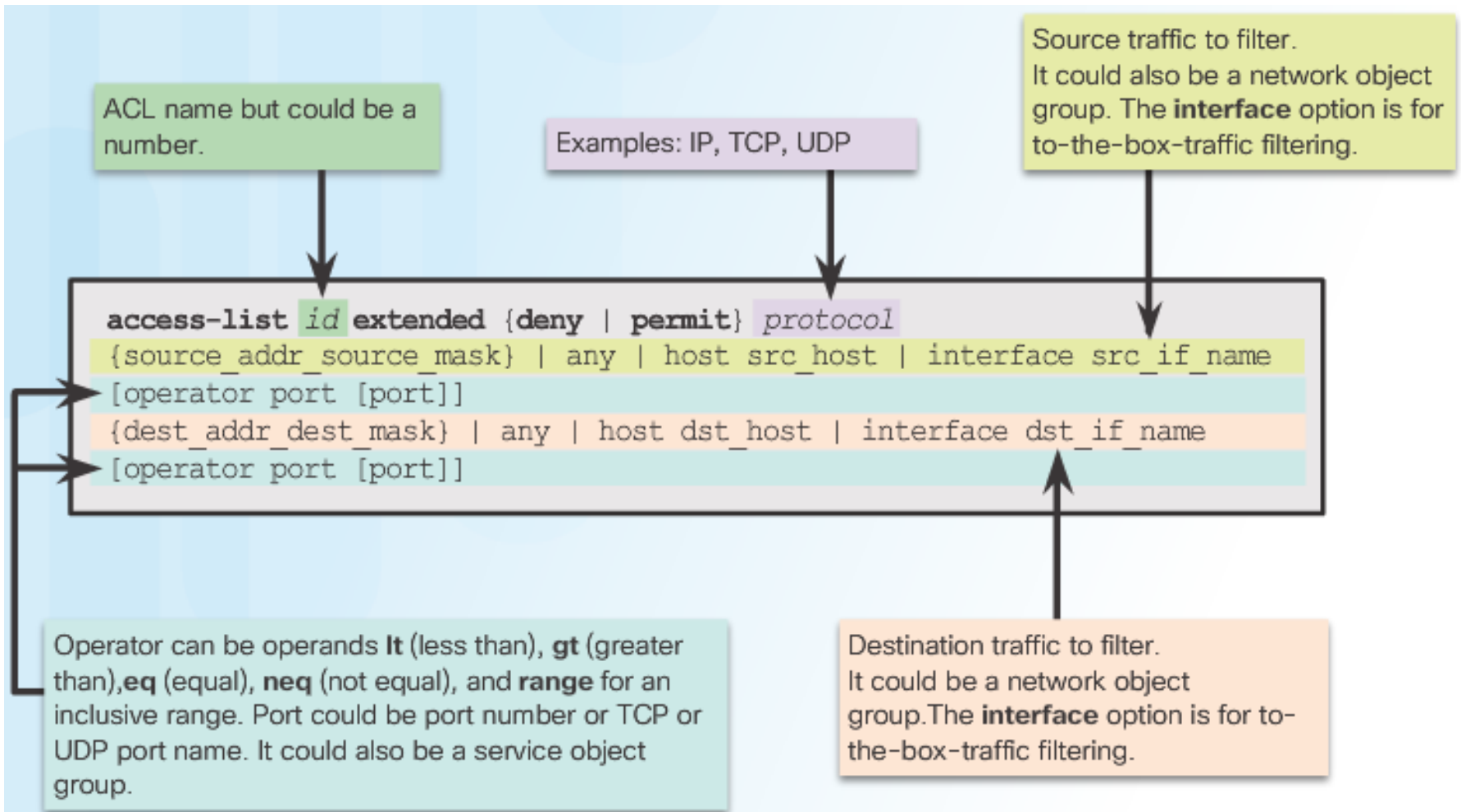
```
Use this to configure policy for IP traffic through the firewall
```

```
[no] access-list <id> [line <line_num>] [extended] {deny | permit}  
    {<protocol> | object-group {<service_obj_grp_id> |  
    <protocol_obj_grp_id>} | object <service_object_name>}  
    [user-group [<domain_nickname>\\]<user_group_name> |  
    user [<domain_nickname>\\<user_name> |  
    object-group-user <object_group_user_name>]  
    [security-group {name <sgname> | tag <sgt>}] |  
    object-group-security <security_obj_grp_id>]  
    {host <sip> | <sip> <smask> | <sip-prefix> |  
    interface <ifc> | any | any4 | any6  
    object-group <network_obj_grp_id> |  
    object <network_obj_name>}  
    [<operator> <port> [<port>] |  
    object-group <service_obj_grp_id>]  
    [security-group {name <sgname> | tag <sgt>}] |  
    object-group-security <security_obj_grp_id>]  
    {host <dip> | <dip> <dmask> | <dip-prefix> |  
    interface <ifc> | any | any4 | any6
```

```
<--- More --->
```

ACL-ek konfigurálása

Kiterjesztett ACL szintaktikája



ACL-ek alkalmazása

access-group parancs szintaktikája

```
access-group id { in | out } interface if_name [ per-user-override | control-plane ]
```

Syntax	Description
access-group	Keyword used to apply an ACL to an interface.
<i>id</i>	The name of the actual ACL to be applied to an interface.
in	The ACL will filter inbound packets.
out	The ACL will filter outbound packets
interface	Keyword to specify the interface to which to apply the ACL.
<i>if_name</i>	The name of the interface to which to apply an ACL.
per-user-override	Option that allows downloadable ACLs to override the entries on the interface ACL.
control-plane	Keyword to specify whether the applied ACL analyzes traffic destined to ASA for management purposes.

ACL-ek alkalmazása példa

```
access-list ACL-IN extended permit ip any any  
access-group ACL-IN in interface inside
```

```
access-list ACL-IN extended deny tcp 192.168.1.0 255.255.255.0 209.165.201.0  
255.255.255.224  
access-list ACL-IN extended permit ip any any  
access-group ACL-IN in interface inside
```

```
access-list ACL-IN extended deny tcp any host 209.165.201.29 eq www  
access-list ACL-IN extended permit ip any any  
access-group ACL-IN in interface inside
```

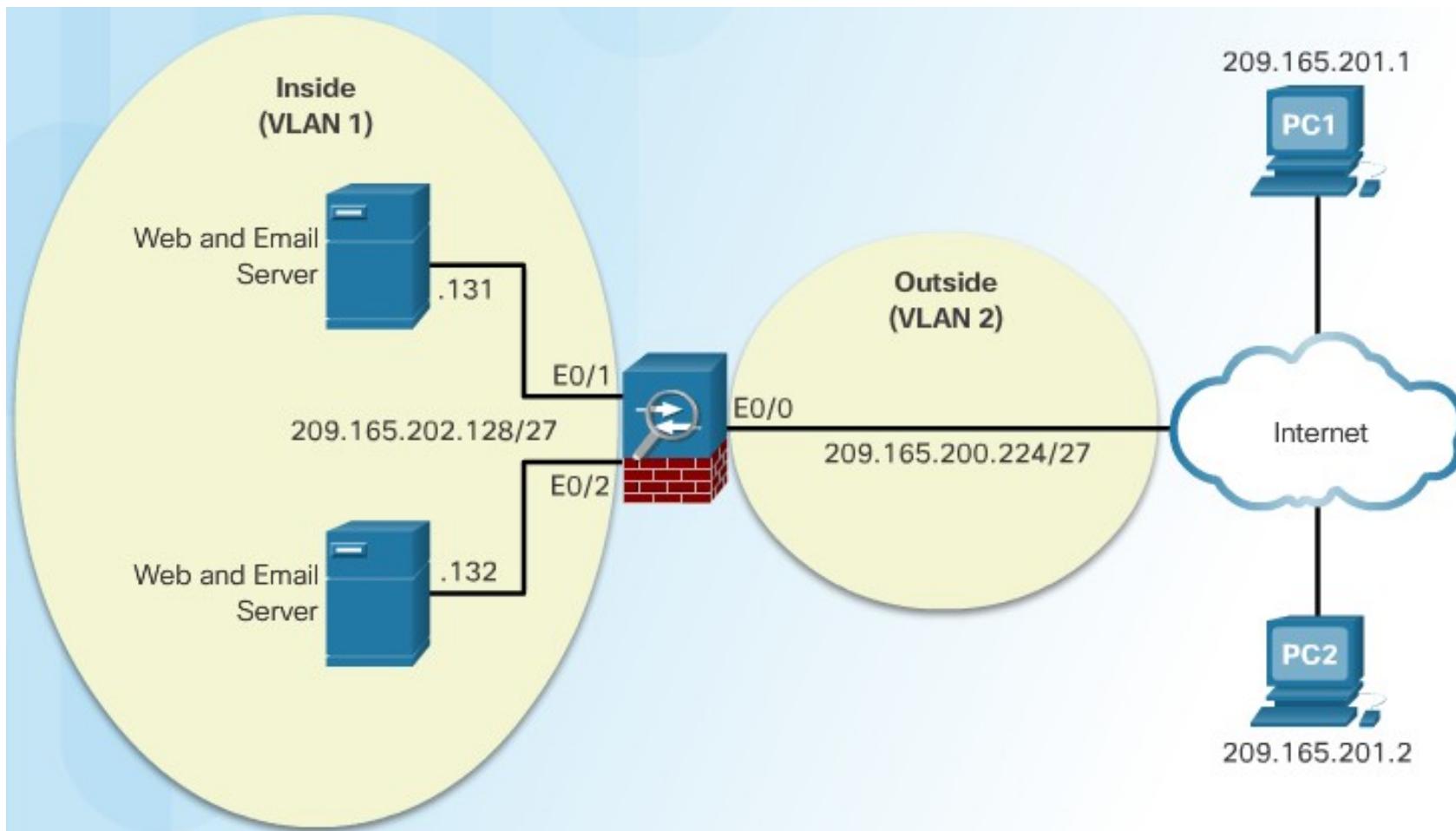
ACL-ek és az objektum csoportok

- Leegyszerűsítik a nehéz ACL-ek megírását. Azzal, hogy a hasonló elemeket csoportosítjuk az ACE-k száma csökken.

```
access-list id extended { deny | permit } protocol object-group  
network-obj-grp-id object-group network-obj-grp-id object-group  
service-obj-grp-id
```

ACL-ek és az objektum csoportok

ACL minta topológia



ACL-ek és az objektum csoportok példa

```
CCNAS-ASA(config)# object-group network NET-HOSTS
CCNAS-ASA(config-network-object-group)# description OG matches PC-A and PC-B
CCNAS-ASA(config-network-object-group)# network-object host 209.165.201.1
CCNAS-ASA(config-network-object-group)# network-object host 209.165.201.2
CCNAS-ASA(config-network-object-group)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# object-group network SERVERS
CCNAS-ASA(config-network-object-group)# description OG matches Web / Email Servers
CCNAS-ASA(config-network-object-group)# network-object host 209.165.202.131
CCNAS-ASA(config-network-object-group)# network-object host 209.165.202.132
CCNAS-ASA(config-network-object-group)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# object-group service HTTP-SMTP tcp
CCNAS-ASA(config-service-object-group)# description OG matches SMTP / WEB traffic
CCNAS-ASA(config-service-object-group)# port-object eq smtp
CCNAS-ASA(config-service-object-group)# port-object eq www
CCNAS-ASA(config-service-object-group)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# access-list ACL-IN remark Only permit PC-A / PC-B -> Internal Servers
CCNAS-ASA(config)# access-list ACL-IN extended permit tcp object-group NET-HOSTS
object-group SERVERS object-group HTTP-SMTP
```

Ellenőrzés:

```
CCNAS-ASA(config)# show running-config access-list
access-list ACL-IN remark Only permit PC-A / PC-B -> Internal Servers
access-list ACL-IN extended permit tcp object-group NET-HOSTS object-group SERVERS
object-group HTTP-SMTP
CCNAS-ASA(config)#
```

9.2.5

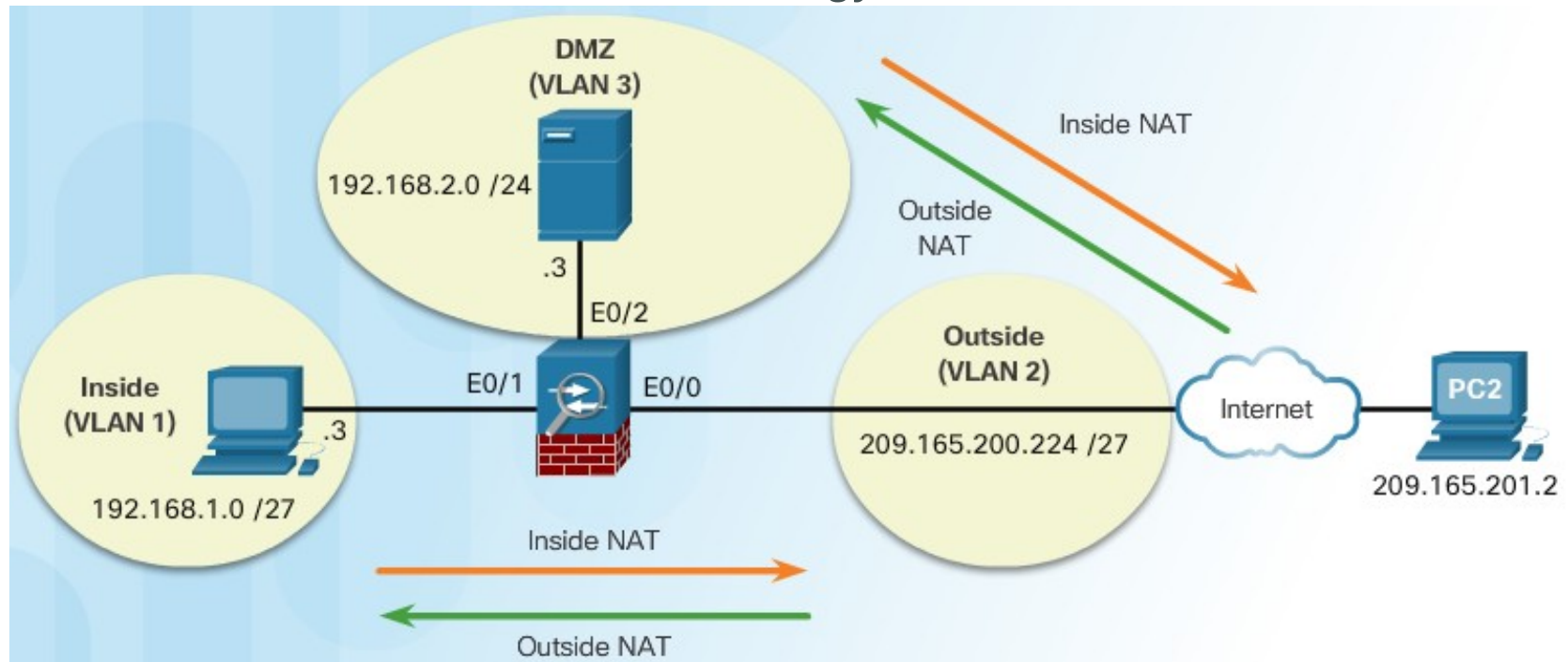
NAT szolgáltatás az ASA-n



ASA NAT áttekintés

NAT megvalósítások típusai:

- Inside NAT privát → publikus
- Outside NAT kintről akarunk elérni egy belső gépet
- Bidirectional NAT az előző kettő együttes alkalmazása



Dinamikus NAT beállítása

Hálózati objektumokkal kell létrehozni!

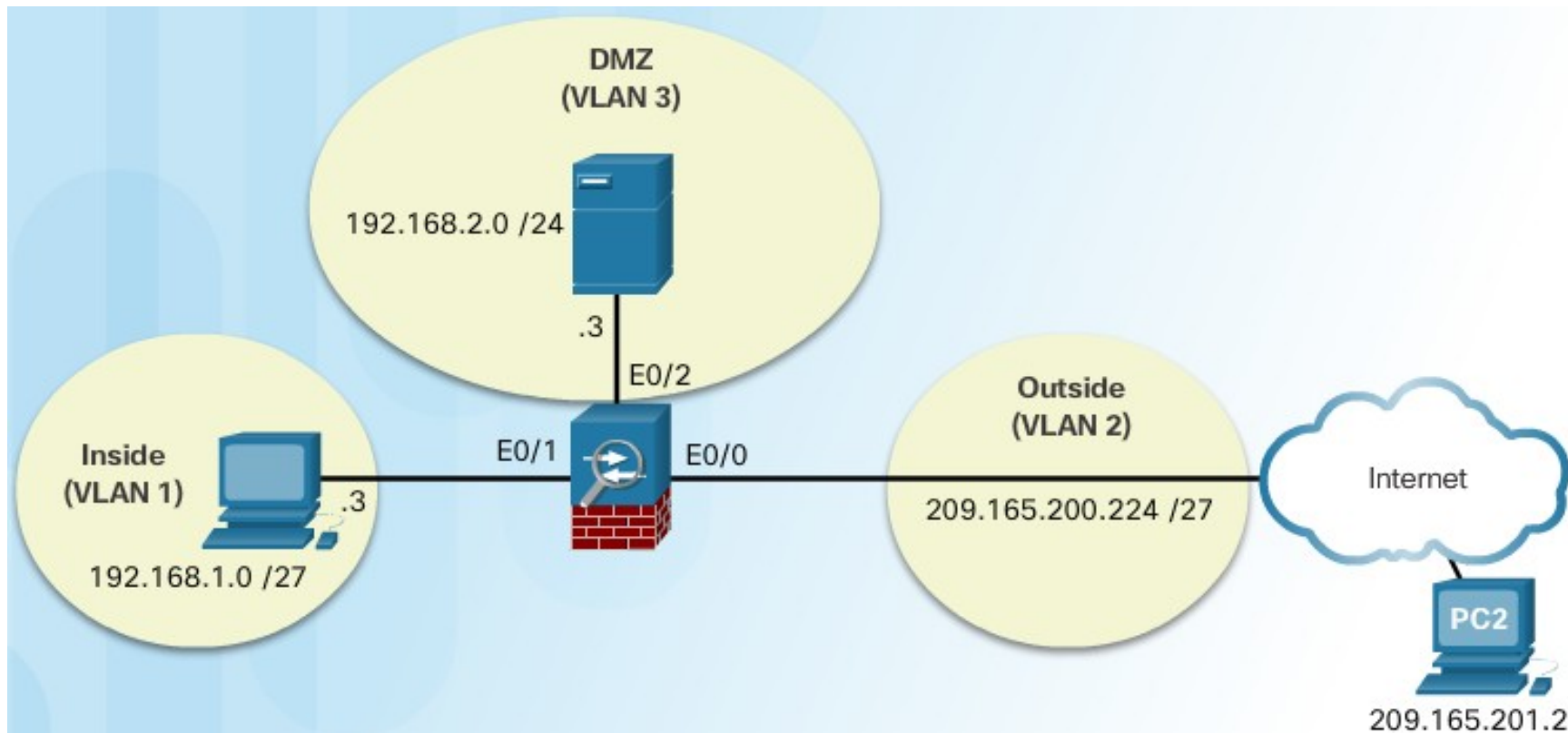
Kell egy hálózati objektum a pool meghatározására.

Kell egy hálózati objektum a belső címek azonosítására és a pool-lal való összekötésére.

A visszajövő forgalmat pedig policy map-pel kell engedélyezni.

Dinamikus NAT beállítása

Minta topológia



Dinamikus NAT beállítása

```
CCNAS-ASA(config)# object network PUBLIC
CCNAS-ASA(config-network-object)# range 209.165.200.240 209.165.200.248
CCNAS-ASA(config-network-object)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# object network DYNAMIC-NAT
CCNAS-ASA(config-network-object)# subnet 192.168.1.0 255.255.255.224
CCNAS-ASA(config-network-object)# nat (inside,outside) dynamic PUBLIC
CCNAS-ASA(config-network-object)# end
CCNAS-ASA#
```

Pool meghatározása.

Belső címek azonosítása és a pool-lal összekötés

```
CCNAS-ASA(config)# policy-map global_policy
CCNAS-ASA(config-pmap)# class inspection_default
CCNAS-ASA(config-cmap)# access-list ICMPACL extended permit icmp any any
CCNAS-ASA(config)# access-group ICMPACL in interface outside
CCNAS-ASA(config)#
```

Visszajövő forgalom engedélyezése

Ellenőrzés

```
CCNAS-ASA(config)# show xlate
1 in use, 1 most used
Flags: D - DNS, e - extended, I - identity, i - dynamic, r - portmap,
       s - static, T - twice, N - net-to-net

NAT from inside:192.168.1.3 to outside:209.165.200.242 flags i idle 0:00:02 timeout 3:00:00
CCNAS-ASA(config)#
CCNAS-ASA(config)# show nat

Auto NAT Policies (Section 2)
1 (inside) to (outside) source dynamic DYNAMIC-NAT PUBLIC
   translate_hits = 1, untranslate_hits = 1
CCNAS-ASA(config)#
CCNAS-ASA(config)# show nat detail

Auto NAT Policies (Section 2)
1 (inside) to (outside) source dynamic DYNAMIC-NAT PUBLIC
   translate_hits = 1, untranslate_hits = 1
   Source - Origin: 192.168.1.0/27, Translated: 209.165.200.240-209.165.200.248
CCNAS-ASA(config)#
```

Dinamikus PAT beállítása

Itt csak a második hálózati objektum kell a dynamic bejegyzéssel

```
CCNAS-ASA(config)# object network INSIDE-NET
CCNAS-ASA(config-network-object)# subnet 192.168.1.0 255.255.255.224
CCNAS-ASA(config-network-object)# nat (inside,outside) dynamic interface
CCNAS-ASA(config-network-object)# end
CCNAS-ASA#
```

Ellenőrzés

```
CCNAS-ASA# show xlate
1 in use, 1 most used
Flags: D - DNS, e - extended, I - identity, i - dynamic, r - portmap,
       s - static, T - twice, N - net-to-net

ICMP PAT from inside:192.168.1.3/1 to outside:209.165.200.226/1 flags ri idle
0:00:02 timeout 0:00:30
CCNAS-ASA#
```

Statikus NAT beállítása

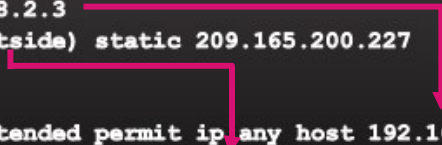
- Itt kell mindhárom lépés. Az első hálózati objektumnál a host opciót kell használni

```
CCNAS-ASA(config)# interface Vlan3
CCNAS-ASA(config-if)# no forward interface Vlan1
CCNAS-ASA(config-if)# nameif dmz
INFO: Security level for "dmz" set to 0 by default.
CCNAS-ASA(config-if)# security-level 70
CCNAS-ASA(config-if)# ip address 192.168.2.1 255.255.255.0
CCNAS-ASA(config-if)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# interface Ethernet0/2
CCNAS-ASA(config-if)# switchport access vlan 3
CCNAS-ASA(config-if)# no shut
CCNAS-ASA(config-if)# exit
CCNAS-ASA(config)#
```

DMZ interfész
beállítás példa

Statikus NAT
beállítás

```
CCNAS-ASA(config)# object network DMZ-SERVER
CCNAS-ASA(config-network-object)# host 192.168.2.3
CCNAS-ASA(config-network-object)# nat (dmz,outside) static 209.165.200.227
CCNAS-ASA(config-network-object)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# access-list OUTSIDE-DMZ extended permit ip any host 192.168.2.3
CCNAS-ASA(config)# access-group OUTSIDE-DMZ in interface outside
CCNAS-ASA(config)#
CCNAS-ASA(config)# policy-map global_policy
CCNAS-ASA(config-pmap)# class inspection default
CCNAS-ASA(config-pmap-c)# access-list ICMPACL extended permit icmp any any
CCNAS-ASA(config)# access-group ICMPACL in interface dmz
CCNAS-ASA(config)#
```



Statikus NAT beállítása

Ellenőrzés:

```
CCNAS-ASA(config)# show xlate
2 in use, 2 most used
Flags: D - DNS, e - extended, I - identity, i - dynamic, r - portmap,
      s - static, T - twice, N - net-to-net
NAT from dmz:192.168.2.3 to outside:209.165.200.227
      flags s idle 0:00:21 timeout 0:00:00

NAT from inside:192.168.1.3 to outside:209.165.200.242 flags i idle 0:09:06 timeout
3:00:00
CCNAS-ASA(config)#
CCNAS-ASA(config)# show nat detail

Auto NAT Policies (Section 2)
1 (dmz) to (outside) source static DMZ-SERVER 209.165.200.227
   translate_hits = 1, untranslate_hits = 1
   Source - Origin: 192.168.2.3/32, Translated: 209.165.200.227/32
2 (inside) to (outside) source dynamic DYNAMIC-NAT PUBLIC
   translate_hits = 1, untranslate_hits = 1
   Source - Origin: 192.168.1.0/27, Translated: 209.165.200.240-209.165.200.248
CCNAS-ASA(config)#
```

9.2.6 AAA



AAA áttekintés

Authentication
Who are you?

Authorization
How much can you spend?

Accounting
What did you spend it on?

Account Number: 1234-567-890
Statement Closing Date: 01-31-01
Current Amount Due: \$278.50

JOE EMPLOYEE
456 SKYVIEW DRIVE
HOMETOWN, USA 99900-1234
872919345 00178255000000003

MAIL PAYMENT TO:
THE BANK
132 VINE STREET
ANYTOWN, USA 67500-0010

Detach here and return upper portion with check or money order. Do not staple or fold.

Statement of Personal Credit Card Account
Retain this portion for your files.

Cardmember Name: JOE EMPLOYEE
Account Number: 1234-456-890
Statement Closing Date: 01-31-01

Statement Date: 02-01-01
Closing Date: 01-31-01
Credit Limit: \$1,500.00
New Balance: \$278.50

Payment Due Date: 03-01-01
Credit Available: \$1221.50
Minimum Payment Due: \$20.00

Account Summary

Previous Balance:	+74.24	Transaction Fees:	+3.00
Purchases:	+250.50	Annual Fees:	+25.00
Cash Advances:	+0	Current Amount Due:	+250.50
Payments:	-74.25	Amount Past Due:	+0
Finance Charge:	+0	Amount Over Credit Line:	+0
Late Charge:	+0	NEW BALANCE:	\$278.50

Reference Number	Sold	Posted	Activity Since Last Statement	Amount
43210987	01-03	01-13	Payment, Thank You	-\$74.25
01234567	01-12	01-13	Wings 'N' Things Anytown, USA	\$25.25
78901234	01-14	01-17	Record Release Anytown, USA	\$40.00
45678901	01-14	01-17	Sports Stadium Anytown, USA	\$75.25
3210987	01-22	01-23	Tie Tack Anytown, USA	\$20.75
76543210	01-29	01-30	Electronic World Anytown, USA	\$89.25
2345678		01-30	Transaction Fees	\$3.00
34567890		01-01	Annual Fee	\$25.00

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Helyi adatbázis és szerverek

RADIUS és TACACS+ szerver parancsai

ASA Command	Description
<code>aaa-server server-tag protocol protocol</code>	Creates a TACACS+ or RADIUS AAA server group.
<code>aaa-server server-tag [(interface-name)] host {server-ip name} [key]</code>	- Configures a AAA server as part of a AAA server group. - Also configures AAA server parameters that are host-specific.

Minta az AAA TACACS+ szervertes konfigurációra

```
CCNAS-ASA(config)# username Admin password class privilege 15
CCNAS-ASA(config)#
CCNAS-ASA(config)# show run username
username Admin password obYXcKAuUW.jT5NE encrypted privilege 15
CCNAS-ASA(config)#
CCNAS-ASA(config)# aaa-server TACACS-SVR protocol tacacs+
CCNAS-ASA(config-aaa-server-group)# aaa-server TACACS-SVR (dmz) host 192.168.2.3
CCNAS-ASA(config-aaa-server-host)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# show run aaa-server
aaa-server TACACS-SVR protocol tacacs+
aaa-server TACACS-SVR (dmz) host 192.168.2.3
key *****
CCNAS-ASA(config)#
```

AAA konfiguráció

```
CCNAS-ASA(config)# aaa authentication http console TACACS-SVR LOCAL
CCNAS-ASA(config)# aaa authentication enable console TACACS-SVR LOCAL
CCNAS-ASA(config)# aaa authentication http console TACACS-SVR LOCAL
CCNAS-ASA(config)# aaa authentication serial console TACACS-SVR LOCAL
CCNAS-ASA(config)# aaa authentication ssh console TACACS-SVR LOCAL
CCNAS-ASA(config)# aaa authentication telnet console TACACS-SVR LOCAL
CCNAS-ASA(config)#
CCNAS-ASA(config)# show run aaa
aaa authentication enable console TACACS-SVR LOCAL
aaa authentication http console TACACS-SVR LOCAL
aaa authentication serial console TACACS-SVR LOCAL
aaa authentication ssh console TACACS-SVR LOCAL
aaa authentication telnet console TACACS-SVR LOCAL
CCNAS-ASA(config)# exit
CCNAS-ASA# disable
CCNAS-ASA> exit
```

Logoff

Username: **Admin**

Password: *********

Type help or '?' for a list of available commands.

CCNAS-ASA>

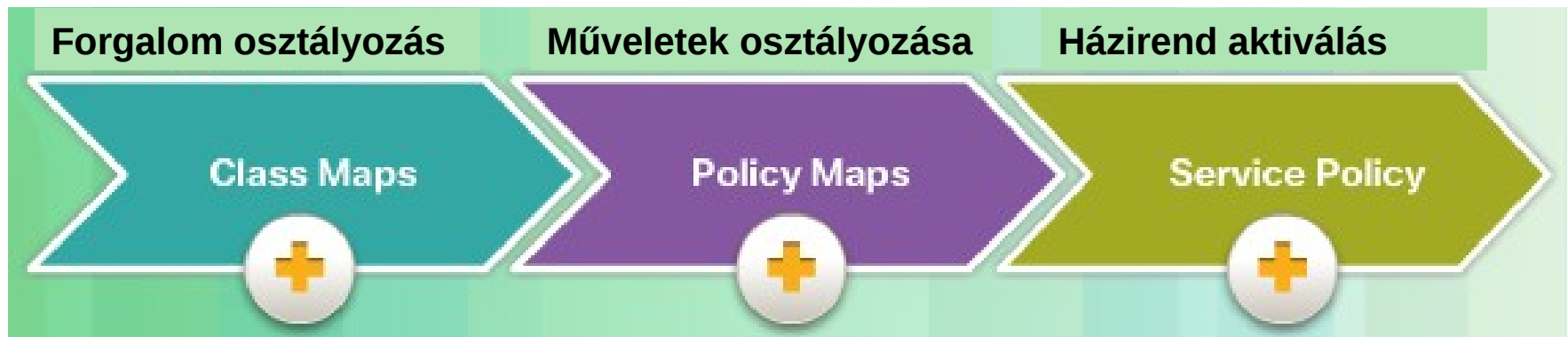
9.2.7

Szolgáltatás házirendek az ASA-n



A Modular Policy Framework áttekintése

- Forgalomfigyelésre, QoS esetén és az ASA-hoz menő forgalom szűrésére használják.
- Moduláris, objektum-orientált, hierarchikus szabályok.



Konfigurálás lépései

- 1. (opcionális) ACL létrehozása, ami az adott forgalmat azonosítja a class-map-ben erre hivatkozhatunk.
- 2. class map létrehozása, ami a forgalmat azonosítja.
- 3. policy map létrehozása, amivel megadható mi történjen.
- 4. service policy létrehozása, amivel a policy map-et az interfészhez kapcsoljuk.

Class Map létrehozása

```
CCNAS-ASA(config)# access-list UDP permit udp any any
CCNAS-ASA(config)# access-list TCP permit tcp any any
CCNAS-ASA(config)# access-list SERVER permit ip any host 10.1.1.1
CCNAS-ASA(config)#
CCNAS-ASA(config)# class-map ALL-TCP
CCNAS-ASA(config-cmap)# description "This class-map matches all TCP traffic"
CCNAS-ASA(config-cmap)# match access-list TCP
CCNAS-ASA(config-cmap)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# class-map ALL-UDP
CCNAS-ASA(config-cmap)# description "This class-map matches all UDP traffic"
CCNAS-ASA(config-cmap)# match access-list UDP
CCNAS-ASA(config-cmap)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# class-map ALL-HTTP
CCNAS-ASA(config-cmap)# description "This class-map matches all HTTP traffic"
CCNAS-ASA(config-cmap)# match port TCP eq http
CCNAS-ASA(config-cmap)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# class-map TO-SERVER
CCNAS-ASA(config-cmap)# description "Class map matches traffic 10.1.1.1"
CCNAS-ASA(config-cmap)# match access-list SERVER
CCNAS-ASA(config-cmap)# exit
CCNAS-ASA(config)#
```

Házirend létrehozása és aktiválása

Példa az MPF konfigurációra

```
CCNAS-ASA(config)# access-list TFTP-TRAFFIC permit udp any any eq 69
CCNAS-ASA(config)#
CCNAS-ASA(config)# class-map CLASS-TFTP
CCNAS-ASA(config-cmap)# match access-list TFTP-TRAFFIC
CCNAS-ASA(config-cmap)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# policy-map POLICY-TFTP
CCNAS-ASA(config-pmap)# class CLASS-TFTP
CCNAS-ASA(config-pmap-c)# inspect tftp
CCNAS-ASA(config-pmap-c)# exit
CCNAS-ASA(config-pmap)# exit
CCNAS-ASA(config)#
CCNAS-ASA(config)# service-policy POLICY-TFTP global
CCNAS-ASA(config)#
```

Az alapértelmezett ASA házirendek

Alapértelmezett Service Policy

<output omitted>

```
class-map inspection_default  
match default-inspection-traffic
```

Class map consists of one statement matching a special keyword **default-inspection-traffic**.

```
policy-map global_policy  
class inspection_default  
inspect dns preset_dns_map  
inspect ftp  
inspect h323 h225  
inspect h323 ras  
inspect ip-options  
inspect netbios  
inspect rsh  
inspect rtsp  
inspect skinny  
inspect esmtp  
inspect sqlnet  
inspect sunrpc  
inspect tftp  
inspect sip  
inspect xdmcp
```

Policy map associates actions to perform on the traffic identified in the class map.

```
service-policy global_policy global
```

Service policy applies a policy map to an interface or to all interfaces using the keyword **global**. The **global** keyword applies a policy map to interfaces that do not have a specific policy applied.

<output omitted>